

IESE CONSULTING CLUB

# CASE BOOK 2021

IN PARTNERSHIP WITH BOSTON CONSULTING GROUP

2<sup>nd</sup> Edition

# FOREWORD

Dear Reader,

In partnership with Boston Consulting Group (BCG), we are proud to share the second edition of The IESE Case Book.

The IESE Consulting Club collaborated with BCG to create a competition for consulting club members at IESE (Class of 2021 and 2022) to write and submit original cases, to help the next generation of aspiring consultants prepare for interviews. Each submitted case passed through a careful and completely blind evaluation process conducted by the IESE Consulting Club and BCG, which selected the 5 best cases based on pre-defined evaluation criteria. These 5 best cases form the IESE Case Book 2021 which will enable the reader to:

- Develop interview guidance techniques to practice peer-to-peer preparation
- Improve problem-solving skills
- Mimic real-life interview situations with solid cases, accompanied with BCG insights

We have paid special attention to the user experience so that the whole process of practicing peer-to-peer mocks becomes easier, seamless, and intuitive. We have tried our best to provide a detailed explanation of how to use the book and some tips for interviewer guidance throughout all cases.

We would like to thank Boston Consulting Group for their partnership and support in bringing this case book to life. Wishing you the best of luck and hope you enjoy your preparation journey!

IESE Consulting Club

# CONTENT

- ❖ **ACKNOWLEDGEMENT**
- ❖ HOW TO USE THE CASE BOOK
- ❖ CASES 2<sup>nd</sup> Edition
- ❖ CASES 1<sup>st</sup> Edition

# Acknowledgement

The second edition of the IESE Case Book is the result of coordinated efforts between Boston Consulting Group and all IESE students that employed time, creativity, and dedication in bringing this case book to life.

The collection of cases that you will find in the following pages were created as part of the 2<sup>nd</sup> IESE Case book Writing Competition amongst consulting club members, who were challenged to write cases based on business problems that could help consulting candidates in their preparation process.

Each one of the cases submitted passed through a careful and completely blind evaluation process, conducted by the IESE Consulting Club and BCG, which selected the five best cases based on pre-defined evaluation criteria.

We would like to thank the BCG Team for their time, support, and guidance in this endeavour, the participants for the time and effort they put to develop their cases, and the IESE Case book Team members from Class of 2021 and 2022 who helped bring this book to life in its final format.

# CONTENT

- ❖ ACKNOWLEDGEMENT
- ❖ **HOW TO USE THE CASE BOOK**
- ❖ CASES 2<sup>nd</sup> Edition
- ❖ CASES 1<sup>st</sup> Edition

# How To Use This Case Book

**1 Eurotech**

**PROMPT**

Two technologic firms within the audiovisual tech industry have decided to join forces. Their market: manufacturing of Hardware components for audio-visual transmission has been declining at a 14% rate and this has affected their profitability jeopardizing their future operations.

Both of their CEOs have agreed on contracting a consulting firm to help them navigate the process.

**Your job is to size the potential synergies that the operation may yield for the NewCo in the future**

(\*) Refers to the manufacturing and installation of reception devices (antennas), transmission infrastructure (Wiring), and end-terminals (set-top-boxes, smart TVs, phones...)

**2 Eurotech – Exhibit 2**

**REVENUES FY20**

| All Numbers in ME                                       | French Company | Polish Company |
|---------------------------------------------------------|----------------|----------------|
| <b>Revenues</b>                                         | <b>70</b>      | <b>85</b>      |
| <b>Variable Costs</b>                                   | <b>10</b>      | <b>11.5</b>    |
| Raw Materials (components, wirings, hardware pieces...) | 9              | 9.5            |
| Transportation                                          | 1              | 1.5            |
| Energy                                                  | 1              | 0.5            |
| <b>Fixed Costs</b>                                      | <b>55</b>      | <b>50.5</b>    |
| Production (Salaries of employees)                      | 42             | 33.5           |
| Research & Development                                  | 10             | 15             |
| Sales and Administration                                | 3              | 2              |
| <b>Operational Profit</b>                               | <b>5</b>       | <b>23</b>      |

**ADDITIONAL INFO**

- Variable Costs** have remained stable over the last 10 years and are not expected to change since relation with the provider is good
- Fixed costs** have been increasing over the last 5 years at a slow steady pace (no hiring nor firing has occurred for the last 6 years)

**3 Eurotech**

**BRAINSTORMING QUESTION**

Last, the interviewer must ask the candidate to elaborate on the last point of the plan, which is risks/feasibility. The candidate must elaborate a list of risks that could harm the synergies and the profitability of the NewCo in the future. (Broader scope)

**SAMPLE ANSWERS**

The candidate should explore the different risks associated with the future of the NewCo. If he proposes a structure that is linked to profitability, it would be considered a plus, i.e.:

| REVENUES                                                                                                                                                                                                        | VARIABLE COSTS                                                                                                                                                                                                                      | FIX COSTS                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Substitute products</li> <li>Competitive environment (i.e., price-war)</li> <li>Failure in the R&amp;D process</li> <li>Harming client relations</li> <li>...</li> </ul> | <ul style="list-style-type: none"> <li>Increase in the price of components</li> <li>Decline in bargaining power against suppliers (i.e., bankruptcy of current providers)</li> <li>Increase in the price of utilities...</li> </ul> | <ul style="list-style-type: none"> <li>Moving production from France to Poland might not be possible (gov. pressure, legal procedures...)</li> <li>Employee dismissal may harm reputation in the French market</li> <li>Loss of capacity in the R&amp;D function...</li> </ul> |

**4** Industry: Easy Case Type: Medium Function: Hard

We designed this book to be practical and straightforward for both the interviewer and the interviewee. Read the following instructions to ensure a smooth application process and to extract most value out of this case book.

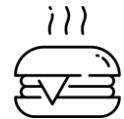
- 1 Red titles** mean information that the interviewer has to give to the interviewee, including the prompt, clarifying questions and exhibits.
- 2 Exhibit pages** provide necessary information to interviewees solve the cases and should be handed in their entirety when instructions asked to do so
- 3 Green titles** mean information that can help the interviewer in guiding the case including expected takeaways, expected considerations, calculations and sample recommendations. Interviewers should not disclose this information to candidates but use it to guide themselves into the flow of the case and help candidates in navigating the numbers.
- 4** Each case is classified by its industry, theme, and concept tested as well as by its level of difficulty.

# CONTENT

- ❖ ACKNOWLEDGEMENT
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- ❖ **CASES 2<sup>nd</sup> Edition**



**Eurotech** *By Íñigo Arteche (IESE MBA 2022)*



**Crunch Yo' Burger** *By Pieter Swart (IESE MBA 2021)*



**Transantiago** *By Michael Stefanic (IESE MBA 2022)*



**California Wildfires** *By Alan Bleiberg (IESE MBA 2022)*



**South bank** *By Giancarlo Young (IESE MBA 2022)*

- ❖ CASES 1<sup>st</sup> Edition

# Eurotech

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*By Íñigo Arteche*



|               |               |
|---------------|---------------|
| TMT           | Easy          |
| M&A           | <b>Medium</b> |
| Profitability | Hard          |

## CASE FLOW – FOR INTERVIEWER

**Total time: 36-45 min**



### Prompt & exhibit I

Time: 5-7 mins

Main Objectives:

- Understand the industry & company size
- Address the problem to be solved: synergy calculation

### Structure

Time: 6-10 min

Main Objectives:

- Elaborate a structure to address the problem
- Use main levers (cost and revenue) as part of the structure

### Exhibit II & quant

Time: 12-15 min

Main Objectives:

- Calculate the main synergies
- Notice one-off costs
- Understand the difference between target and potential synergy

### Risk brainstorming

Time: 8 mins

Main Objectives:

- Elaborate a list of risks associated with the NewCo synergy implementation and future operations

### Recommendation

Time: 5 mins

Main Objectives:

- Outline target synergy
- Comment on main synergies
- Elaborate on risks and mitigation actions

(\*) Refers to the manufacturing and installation of reception devices (antennas), transmission infrastructure (Wiring), and end-terminals (set-top-boxes, smart TVs, phones...)

## PROMPT

---

Two technologic firms within the audio-visual tech industry have decided to join forces. Their market\*: manufacturing of Hardware components for audio-visual transmission has been declining at a 14% rate and this has affected their profitability jeopardizing their future operations.

Both of their CEOs have agreed on contracting a consulting firm to help them navigate the process.

**Your job is to size the potential synergies that the operation may yield for the NewCo in the future**

## CLARIFYING POINTS (if asked)

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- If the interviewee asks to know more about the audio-visual industry, show exhibit I
- The companies work on a per project basis, mainly for hospitality segment (75% of their revenue)
- Their operations are limited to the manufacturing & assembly of the Hardware components, the installation is done by a third party
- The companies are not based in the same country, one company is in France and the other in Poland
- They are both of similar size and have similar operations
- The French company owns two plants (one in Southern France and the other next to Paris) while the Polish part owns one bigger plant near Warsaw
- The objective is to improve their common profitability by a 10% of the aggregated revenues

## CASE GUIDANCE

---

The objective of this case is to test the candidate's ability to navigate through an M&A operation between technological firms. It is designed to test brainstorming and focuses on a not very known industry to test the candidate's capacity of applying a structured approach to divide the problem into more manageable parts.

The case is not to test the specific knowledge on the AV infrastructure market, but to test the proficiency on how to confront a merger operation, more concretely, how to approach the synergy calculation.

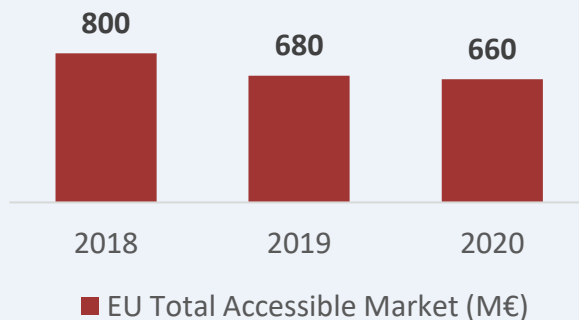
(\*) Refers to the manufacturing and installation of reception devices (antennas), transmission infrastructure (Wiring), and end-terminals (set-top-boxes, smart TVs, phones...)

# Eurotech – Exhibit 1

TMT — Easy  
M&A — Medium  
Profitability — Hard

## INDUSTRY INSIGHTS

- **80% of TAM<sup>1</sup>** depends on Hospitality purchases
- Hotels have recently been moving to **Software services**, reducing dependence on **Hardware providers**
- Market size for HW companies has been declining for the last 3Y (see graph)



## PRODUCT CATEGORIES – HW TRANSMISSION



### RECEPTION

Antennas, Encoders, & Converters\*



### TRANSMISSION

Wiring for different signals and network devices



### TERMINALS

End user devices such as Smart TVs, Tablets, computers...

## MAIN COMPETITORS AND 3Y CAGR

|               | CLIENT |      |       | CLIENT |      |
|---------------|--------|------|-------|--------|------|
| Country       |        |      |       |        |      |
| LY Revenue M€ | 150    | 85   | 83    | 75     | 70   |
| 3Y CAGR       | -20%   | -10% | -5%** | -12.5% | -15% |

Notes: (\*) An encoder/converter is a device that converts different signals (Satellite, Terrestrial, IP...) into playable content (video and/or audio). (\*\*) The Swedish competitor has recently acquired a SW company which is offsetting its decline in sales. | Glossary: (1) Total Accessible Market

# Eurotech – Exhibit 1

---

TMT ..... Easy  
M&A ..... Medium  
Profitability ..... Hard

## GUIDANCE

---

This exhibit is for the candidate to understand better the industry, it is important to try to link it to the reason behind the M&A operation

The candidate should **notice that**:

1. The market is experiencing an overall decline (  $(660-800)/800 = -140/800 = -7/40 = -17.5\%$ ). If the candidate makes the calculation, it would be considered positive
2. This decline is caused by the fact that clients are switching traditional Hardware for Software services. If the candidate notices that the Swedish sales have declined less due to the acquisition of a Software firm it would be considered a plus
3. The operation will create the biggest company in the market ( $75+80 = 155\text{M€}$ )
4. The Polish company initially presents a better financial outlook (lesser decline than most competitors and bigger revenues)
5. The market relies heavily on one type of client: hospitality. If the candidate links the fact that consumers now bring the content linked to their devices (i.e. Netflix, Hulu...) to the increased importance of software services it would be considered a good understanding of the industry behaviour
6. Finally, the candidate should try to guess that since the industry is experiencing such a decrease, the operation might help capture synergies that help both companies survive in such a demanding market

If the candidate does not suggest the reason for the operation, the interviewer might ask him/her to brainstorm a set of possible reasons. **When synergies come up, the interviewer might suggest moving on to the structure.**

## STRUCTURE QUESTION

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The interviewer will ask the candidate to create and present a structure to address the potential synergies upcoming from the M&A operation

## STRUCTURE GUIDANCE

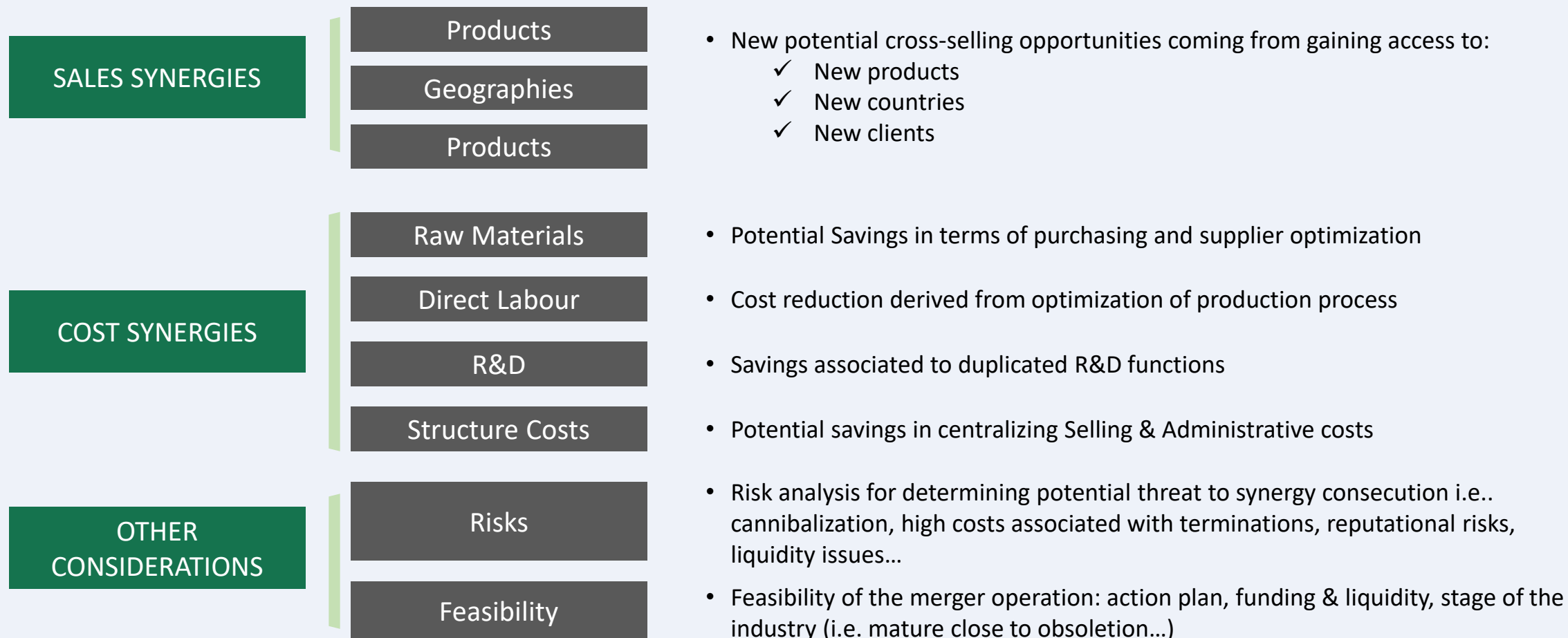
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The candidate needs to identify the information that will help him solve the problem:

- He should indicate that there are two main sources of synergies: sales and costs
  - In terms of sales synergies, the candidate should indicate that there is potential due to access to new clients, countries, and products that were exclusive to each of the companies
  - In terms of cost synergies, the candidate should explain that there are potential synergies and elaborate on the different lines where the companies can save costs from merging (i.e.. Purchase of raw materials, direct labour reductions and/or relocations, R&D costs, and selling/administrative)
  - Last, the candidate could create a bucket to assess the feasibility/risks associated with the synergy calculation

# Eurotech – Proposed structure

TMT ..... Easy  
M&A ..... **Medium**  
Profitability ..... Hard



Notes: (\*) An encoder/converter is a device that converts different signals (Satellite, Terrestrial, IP...) into playable content (video and/or audio). (\*\*) The Swedish competitor has recently acquired a SW company which is offsetting its decline in sales. | Glossary: (1) Total Accessible Market

# Eurotech – Exhibit 2

TMT ..... Easy  
M&A ..... Medium  
Profitability ..... Hard

## REVENUES FY20

| <i>All Numbers in M€</i>                                | <b>French Company</b> | <b>Polish Company</b> |
|---------------------------------------------------------|-----------------------|-----------------------|
| <b>Revenues</b>                                         | <b>70</b>             | <b>85</b>             |
| <b>Variable Costs</b>                                   | <b>10</b>             | <b>11.5</b>           |
| Raw Materials (components, wirings, hardware pieces...) | 9                     | 9.5                   |
| Transportation                                          | 1                     | 1.5                   |
| Energy                                                  | 1                     | 0.5                   |
| <b>Fixed Costs</b>                                      | <b>55</b>             | <b>50.5</b>           |
| Production (Salaries of employees)                      | 42                    | 33.5                  |
| Research & Development                                  | 10                    | 15                    |
| Sales and Administration                                | 3                     | 2                     |
| <b>Operational Profit</b>                               | <b>5</b>              | <b>23</b>             |

## ADDITIONAL INFO

- **Variable Costs** have remained stable over the last 10 years and are not expected to change since relation with the provider is good
- **Fixed costs** have been increasing over the last 5 years at a slow steady pace (no hiring nor firing has occurred for the last 6 years)

# Eurotech – Exhibit 2

---

TMT ..... Easy  
M&A ..... **Medium**  
Profitability ..... Hard

## GUIDANCE

---

### If the candidate asks:

1. There is enough capacity in Poland to move production from France, with the associated increase in Fix Cost (salaries)
2. Termination costs in France (associated with production employees in France) have been estimated to be 10M€ (one-off)
3. The sale of the French Plant would lead to a net profit of 2M€
4. The candidate can consider a target of 70% consecution of the potential synergy. The candidate should indicate that 100% of the synergy is not attainable.

### The candidate should **notice that:**

1. Raw Materials (small electronic components that both companies assemble to create their products) are being managed more efficiently in Poland, it could be an effect of volume but with such a difference it is more likely that they are working with different providers
2. French salaries are consuming a big portion of profitability. The candidate should be able to associate this to the fact that wages in France are much higher than in Poland, and ask if it is possible to shift production from France to Poland
3. He should also notice that Poland is spending more on R&D and state that for a technologic company this is a relevant issue, if he links it with the lesser decline seen in Exhibit I it would be considered a plus

In conclusion, candidate should state that the Polish have a leaner operation and that this gives them the possibility to invest more in R&D and retain sales with more advanced products.

**Gathering all this information the candidate should propose to move on with the structure and start the synergies analysis.**

## QUANTITATIVE ANALYSIS

For the quantitative analysis, the interviewer will ask the candidate to calculate the synergies between the two players using the information contained in the first exhibit.  
(It is important that the candidate makes the calculation with at least 2 decimals, specially for the ratios)

## SOLUTION

- Sales synergies (0M€): here the candidate should brainstorm some ideas on where he thinks the synergies are (i.e. cross-selling of non-common products, access to new countries...). He should state that this type of synergy is more uncertain and speak about the possibility of cannibalization. The interviewer should indicate the candidate to consider zero in terms of sales synergies.
- Raw Materials: here the candidate should use the purchasing ratio from Poland to quantify the potential synergy behind Raw Materials.  $9 - (70 \times 9.5/85) = 1.17\text{M€}$
- Production: in terms of salaries, the candidate should try to explore the possibility of moving production to Poland, as it is more cost-efficient this would yield:  $42 - (70 \times 33.5 / 85) = 42 - 27.588 = 14.41 \text{ M€}$
- Research & Development: here the candidate should notice that this cost is duplicated, with one R&D department they would be able to serve the whole organization, and since the Polish one seems to be more effective (in terms of sales) he could propose to move all R&D to Poland maintaining the Polish budget. **10M€, all French R&D cost**
- Sales & Administration: same reasoning as before yet more unclear since Spain would need to retain some Financial controllers/administrative staff. **2-3M€**

### Target Recurrent Saving of:

$$(1.17 + 14.41 + 10 + 2.5) \times 0.7 = 28.09 \times 0.7 = 19.66 \text{ M€}$$

### One-off Profits/Losses:

$$+2 - 10 = -8 \text{ M€}$$

## BRAINSTORMING QUESTION

Last, the interviewer must ask the candidate to elaborate on the last point of the plan, which is risks/feasibility. The candidate must elaborate a list of risks that could harm the synergies and the profitability of the NewCo in the future. (Broader scope)

## SAMPLE ANSWERS

The candidate should explore the different risks associated with the future of the NewCo. If he proposes a structure that is linked to profitability, it would be considered a plus, i.e.:

### REVENUES

- Substitute products
- Competitive environment (i.e., price-war)
- Failure in the R&D process
- Harming client relations
- ...

### VARIABLE COSTS

- Increase in the price of components
- Decline in bargaining power against suppliers (i.e., bankruptcy of current providers)
- Increase in the price of utilities...

### FIX COSTS

- Moving production from France to Poland might not be possible (gov. pressure, legal procedures...)
- Employee dismissal may harm reputation in the French market
- Loss of capacity in the R&D function...

## RECOMMENDATION QUESTION

---

After this analysis, the client has called us, they want us to have a call with them and propose them our main conclusions over the synergy analysis. You have 2 minutes to elaborate an adequate response that captures what we have been discussing during the project.

## SAMPLE RECOMMENDATION

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- The response should start by first answering the question of how much potential does the operation has in terms of synergies ( ~20M€).
- The candidate should explain that this number has been calculated using a safety coefficient that assumes that the NewCo will be able to capture a 70% of the potential synergy
- Then he should move to explain where the synergies are mainly being captured: Production and R&D
- He should talk as well about one-off profit/losses: in Year 0 (2021) the company would experience a -8M€ loss due mainly to the termination costs associated with the employees in the French plant
- Present a brief risk analysis, if he speaks about the French employment laws and the fact that moving production to Poland might be difficult/costly it would also be considered a plus. Presenting risk mitigation initiatives would also be considered positive.
- Last, he should try to establish a small action plan or suggest a follow up to the call

# Crunch Yo' Burger

---

*By Pieter Swart*



|               |               |
|---------------|---------------|
| Food service  | Easy          |
| Profitability | <b>Medium</b> |
| Operations    | Hard          |

# Crunch Yo' Burger

---

|               |               |
|---------------|---------------|
| Food service  | Easy          |
| Profitability | <b>Medium</b> |
| Operations    | Hard          |

## UPFRONT NOTE FOR INTERVIEWER

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- This case is designed to be as long or short as you want
- There are two optional estimations to test candidate's market sizing ability
  - # Subway restaurants in the US
  - # of sit-in customers per Subway per day
- If you are short on time, you can simply give the candidates the values of these inputs
- If the candidate would like to practice estimations, they can estimate these inputs

# Crunch Yo' Burger

---

Food service ..... Easy  
Profitability ..... **Medium**  
Operations ..... Hard

## INTRO PROMPT

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- You are the CEO of a large multinational fast food chain
- In recent years, you have noticed that your profitability in the US has been lagging behind competitors
- Your Board of Directors would like to know:
  - Why profitability is below that of competitors
  - How you plan to get profitability back in line

## CLARIFYING POINTS (if asked)

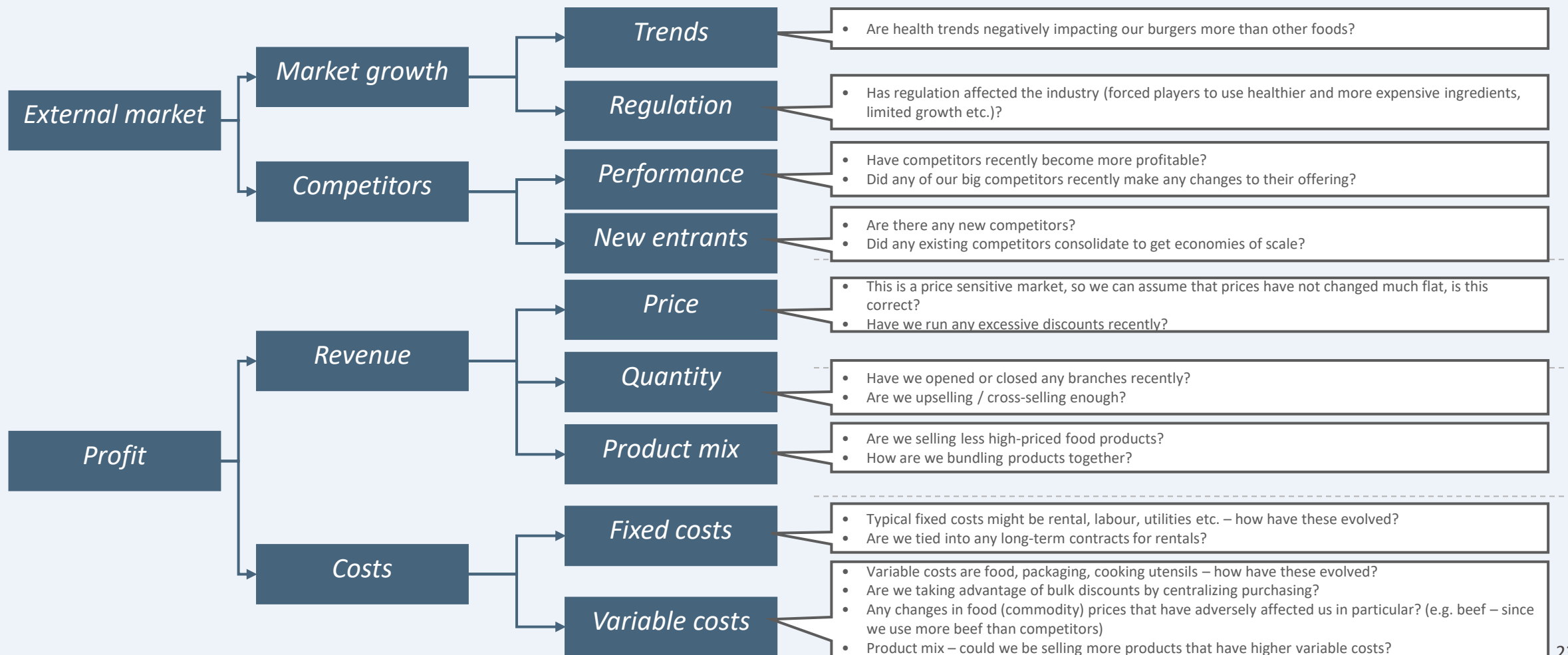
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- Your company sells fast food that is cooked onsite (similar price point to McDonald's/Subway)
- Crunch Yo' Burger operates all its own stores (i.e. no franchises)
- There are four major players in the market, differentiated only by the type of food they offer (prices are the same)
  - Crunch Yo' Burger makes hamburgers
  - TacoCo sells tacos and other Mexican food
  - NoodleCo is focused on different varieties of noodles
  - PizzaCo sells pizzas
- We are concerned only with the US operations of Crunch Yo' Burger and its competitors
- Our customers can be either takeout or eat-in customers
- ***Note for interviewer:*** If asked for detailed figures on profit or revenue, tell the candidate you will show an exhibit after you have seen their structure

# Crunch Yo' Burger

Food service Profitability Operations } Easy  
Medium  
Hard

## EXAMPLE CANDIDATE RESPONSE: SAMPLE STRUCTURE

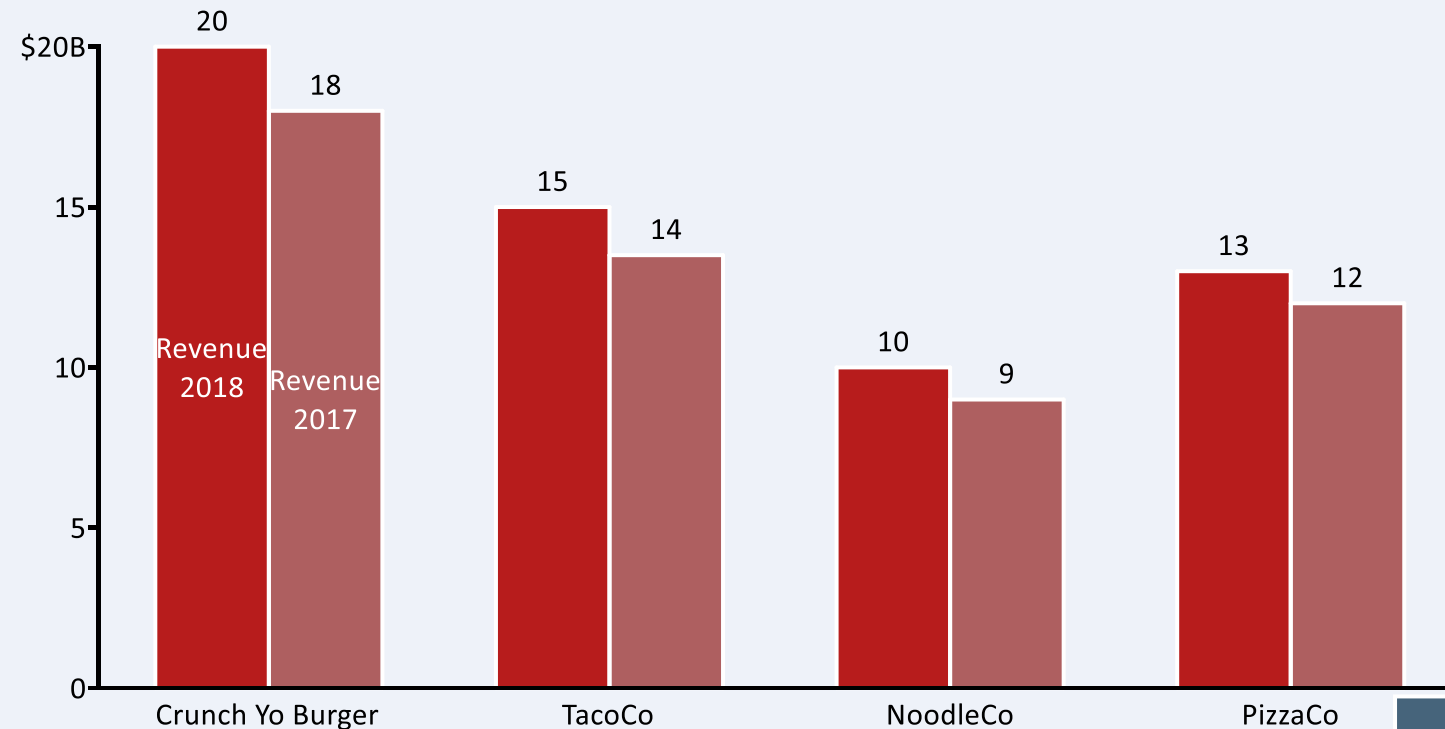


# Crunch Yo' Burger – Exhibit 1

## REVENUE AND PROFITS (ONLY FOR US)

Food service Profitability Operations

Easy  
Medium  
Hard



|                                              |         |         |         |         |
|----------------------------------------------|---------|---------|---------|---------|
| Net profit (2018)                            | \$0.20B | \$0.23B | \$0.15B | \$0.23B |
| Food variable costs as % of sales (2018)     | 25%     | 25%     | 25%     | 25%     |
| Non-food variable costs as % of sales (2018) | 11%     | 10%     | 10%     | 10%     |

Note: Food variable costs relate to raw ingredients that are actually eaten by customers, non-food variable costs are all other variable costs

# Crunch Yo' Burger – Exhibit 1

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Food service ..... Easy  
Profitability ..... Medium  
Operations ..... Hard

## EXAMPLE CANDIDATE RESPONSE

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- This chart shows 2017 and 2018 revenues of the four main players, including our own company (Crunch Yo' Burger), as well as 2018 net profits
- We can see that Crunch Yo' Burger has the highest revenue, thus we are the market leader in fast food in the US
- Revenue growth rates for all competitors have been ~10% since 2017; hypothesis is that this is not a revenue issue
- If we divide net profit by revenues to be able to compare margins, Crunch Yo' Burger has the lowest net profit margin out of its peer group (~1% vs competitors in range of 1.5-2%); this extra 0.5% represents ~\$100M
- This issue is caused by non-food variable costs (11% of sales for Crunch Yo' Burger vs 10% of sales for our peers), and we should investigate these costs further
- ***Note for interviewer:*** The presentation of the chart (showing net profit in absolute terms and variable cost as % of sales in relative terms) is intentional in order to ensure that the candidate can turn these into comparable figures

# Crunch Yo' Burger

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Food service ..... Easy  
Profitability ..... **Medium**  
Operations ..... Hard

## FOLLOW UP QUESTION AFTER EXHIBIT

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- We have concluded that Crunch Yo' Burger's cost issues lie in its non-food related variable costs
- What are some creative ways in which Crunch Yo' Burger could decrease these costs?

## EXAMPLE CANDIDATE RESPONSE

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- Some of the major non-food costs that we would have are:
  - Condiments (ketchup packets or ketchup dispenser)
  - Packaging (burger wrappers, soda cups)
  - Other consumables (napkins toilet paper, plastic cutlery, hand sanitizer)
- Some ways to reduce these could be:
  - Limit ketchup to 1 packet per order, and charge extra for extra packets
  - Use cheaper packaging (although this might pose an environmental and customer experience risk)
  - Limit the number of napkins per customer by handing them out at the teller (this might cause a delay in getting orders out) or by setting up a napkin dispensing machine
- **Note for interviewer:** Try steer the candidate towards napkins, in order to set up the next part of the case; if the candidate does not arrive at this, mention it as a high potential solution

# Crunch Yo' Burger

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Food service ..... Easy  
Profitability ..... **Medium**  
Operations ..... Hard

## CALCULATION QUESTION

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- Crunch Yo' Burger is considering purchasing a machine that dispenses napkins one by one (Napkins are currently placed a large stack, customers take a few at a time before they sit down to eat)
- 1 machine will be needed per store, and needs to be replaced every 2 years; cost is \$1000 per machine
- How much money can Crunch Yo' Burger save per year by implementing this machine?
  - What are the inputs you would need in the calculation and how would you structure it?
  - How much money is saved?
  - This solution will only apply to eat-in customers
- ***Note for interviewer:*** You should try to assess the candidate on two elements here:
  - How well they structure their calculation – do they identify all the elements in a logical way?
  - How well they execute the calculation

# Crunch Yo' Burger

## CALCULATION SOLUTION

Food service Profitability Operations } Easy Medium Hard

Note for interviewer: remember this case has 2 optional estimations! If you want the candidate to estimate, use 'long version' prompts

|                                                                                                                                                                       |   |                                                                                                                                                                                                                                             |   |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                   |   |                                                                                                                                                                                                                             |   |                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------|
| Decrease in napkins per customer                                                                                                                                      | X | Number of customers per store per day                                                                                                                                                                                                       | X | Number of stores in the US                                                                                                                                                                                              | X | Cost per napkin                                                                                                                                                                                   | X | Days per year                                                                                                                                                                                                               | = | Total gross saving                                                                                                    |
| 3                                                                                                                                                                     |   | 960                                                                                                                                                                                                                                         |   | 22500                                                                                                                                                                                                                   |   | \$0.005                                                                                                                                                                                           |   | 365                                                                                                                                                                                                                         |   | ~\$118M                                                                                                               |
| <ul style="list-style-type: none"><li>Before the machine, customers took 5 napkins each</li><li>If machine is implemented, it will reduce to 2 per customer</li></ul> |   | <ul style="list-style-type: none"><li><b>Short version of the case:</b> Give figure of 960 sit-in customers per store per day</li><li><b>Long version of the case:</b> Candidate should estimate this (see Optional Estimation 1)</li></ul> |   | <ul style="list-style-type: none"><li><b>Short version of the case:</b> Give figure of 22500 stores in US</li><li><b>Long version of the case:</b> Candidate should estimate this (see Optional Estimation 2)</li></ul> |   | <ul style="list-style-type: none"><li>Napkins are purchased in boxes of 6000 napkins. A box costs \$28</li><li><b>Note:</b> Allow candidate to round to \$0.005 (half a cent) if needed</li></ul> |   | <ul style="list-style-type: none"><li>Candidate can assume 360 to make calculations easier</li><li>Important that the candidate notes takes note that fast food restaurants tend to be open most days of the year</li></ul> |   | <div>-</div> <div>Cost of machine</div> <div>~\$11M</div> <div>=</div> <div>Total net saving</div> <div>~\$106M</div> |

# Crunch Yo' Burger

Food service ..... Easy  
Profitability ..... **Medium**  
Operations ..... Hard

## OPTIONAL ESTIMATIONS

### EXAMPLE RESPONSE: ESTIMATION #1 – # OF SIT-IN CUSTOMERS IN TYPICAL SUBWAY RESTAURANT PER DAY

- Assume a typical Subway or similar takeout restaurant has ~30 tables, with ~4 chairs per table; capacity of 120 sit-in customers
- Assume it is typically open from 6am to 10pm to sit-in customers (16 hours a day)
- 50% of the time is peak (8/16 hours), 50% of the time is off-peak (8/16 hours), (example breakdown in table below)
- During peak hours, the restaurant is 80% full
- During non-peak hours, the restaurant is 20% full
- Average meal duration is 1 hour (i.e. in a 3 hour time period, a table is rotated 3 times)

| Timeslot    | Peak/ off-peak | Capacity utilisation | Calculation                | # customers |
|-------------|----------------|----------------------|----------------------------|-------------|
| 06:00-09:00 | Peak           | 80%                  | $120 * 3\text{hrs} * 80\%$ | 288         |
| 09:00-12:00 | Off-peak       | 20%                  | $120 * 3\text{hrs} * 20\%$ | 72          |
| 12:00-14:00 | Peak           | 80%                  | $120 * 2\text{hrs} * 80\%$ | 192         |
| 14:00-18:00 | Off-peak       | 20%                  | $120 * 4\text{hrs} * 20\%$ | 96          |
| 18:00-21:00 | Peak           | 80%                  | $120 * 3\text{hrs} * 80\%$ | 288         |
| 21:00-22:00 | Off-peak       | 20%                  | $120 * 1\text{hrs} * 20\%$ | 24          |
|             |                |                      | <b>Total</b>               | <b>960</b>  |

IESE CONSULTING CLUB

### EXAMPLE RESPONSE: ESTIMATION #2 – # OF SUBWAY RESTAURANTS IN THE US

- We have 50 states in the US, let's assume each state is split into rural and urban
  - Urban areas mostly comprise two large cities
  - Rural areas are small towns
- There are 150 Subways stores per city, potential way to estimate this:
  - Each large city is roughly 10 km long by 10km wide
  - Each km has roughly 12 blocks
  - Thus 120 blocks long by 120 blocks wide = 15000 blocks per city
  - Let's assume that as you walk, you are likely to encounter a Subway store every 10 blocks, thus 150 Subways per city
- Let's assume the cities contain 2/3 of all Subway restaurants, and the towns the remaining 1/3, Thus  $2 * 150 = 300$  Subway restaurants in cities, and 150 in towns = 450 Subway restaurants per state
- $450 * 50 \text{ states} = 22,500$  Subway restaurants in the US
- **Note for interviewer:** *There are many potential ways in which the candidate could do this estimation (land area or Subway restaurant per person)*

# Crunch Yo' Burger

## EXAMPLE CANDIDATE RESPONSE: INTERPRETATION OF SAVINGS VALUE

- We have calculated that we will save \$106M per year
- This \$106M represents an additional 0.5% of net profit margin
- With this additional saving, our profit margin will increase from 1% to 1.5%, putting us within the band of our competitors (although still at the lower end)
- We should consider other cost-saving initiatives to continue to increase profitability

## PROMPT: BRAINSTORM

What are some other ways we could save on variable costs?

## EXAMPLE CANDIDATE RESPONSE

- Raw materials bulk discounts
- Extend useful life of consumable utensils
- Run similar initiatives with ketchup, straws etc.
- Decrease cost of napkins (cheaper napkins)
- **Note for interviewer:** *There are many potential options here, extra points for creativity*

# Crunch Yo' Burger

## PROMPT FOR FINAL RECOMMENDATION

- You have a 45-second slot in the Board meeting tomorrow to update the Board on your progress. What will you say?

## EXAMPLE CANDIDATE RESPONSE: RECOMMENDATION

- *(Candidate should assume an air of authority and confidence when addressing board)*
- Good day and welcome
- We have run a diagnosis and can conclude that our lower profitability is caused by high non-food variable costs, specifically in napkins – customers are currently using excessive napkins when they enter our stores
- We have found a way of bringing our net profit margin closer in line with our competitors, increasing it from 1% to 1.5% by implementing a new napkin dispensing machine in all of our stores
- This machine will reduce napkins taken per customer by 3 napkins (60%), and on a national scale, save Crunch Yo' Burger ~\$100M per year
- There are some implementation risks related to customer experience and ease-of-use of this new machine
- In order to mitigate these risks we will pilot the machine in 100 stores nation wide and record feedback and improvements
- Furthermore, we are investigating similar cost-saving initiatives in our ketchup and straw dispensing, in order to surpass the profitability of competitors

# Transantiago

---

By *Michael Stefanic*



|                |               |
|----------------|---------------|
| Transportation | Easy          |
| Profitability  | <b>Medium</b> |
| Public Sector  | Hard          |

# Transantiago

---

Transportation ..... Easy  
Profitability ..... **Medium**  
Public Sector ..... Hard

## PROMPT

---

Transantiago is the public transportation system of Santiago de Chile. It is managed by the Ministry of Transportation and has more than 5 million users, integrating all the city's urban buses and subway.

Transantiago's main source of revenue is the one generated by the tickets sold. They have two types of tickets. One normal ticket, that has a price of 2 dollars per trip, and one discounted tickets, that has a price of 1 dollar.

With this revenue, plus a fixed subsidy of 900 Million USD, Transantiago must cover all their operating costs of the year.

Suppose it is April 2020. The Ministry of Transportation has called you because, After Covid-19, Transantiago has been experiencing a substantial decrease in the demand and is worried if they to cover all their operating costs. **That's why they want you to investigate all the possible impacts of Covid-19 and see how they will be able to cover all their operating costs this year.**

## CLARIFYING POINTS (if asked)

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- The main goal is to see if they will be able to cover all their operating costs this year
- The only actual source of revenue is the transportation tickets sold
- Each ticket can be used in buses and subway or mixing both
- The only price discrimination is if it is a student or not

## CASE GUIDANCE

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This is an interviewee-led case, where the candidate is expected to drive the case and suggest the next course of action.

This case strongly focuses on profitability and requires the interviewee to think about the cost and revenue and analyse the total impact generated considering both. Also, the candidate must consider the risk behind the forecast and the consequences if some of the forecasted numbers are not reached.

From the quantitative aspect, the interviewee will have to connect numbers given in different stages of the case.

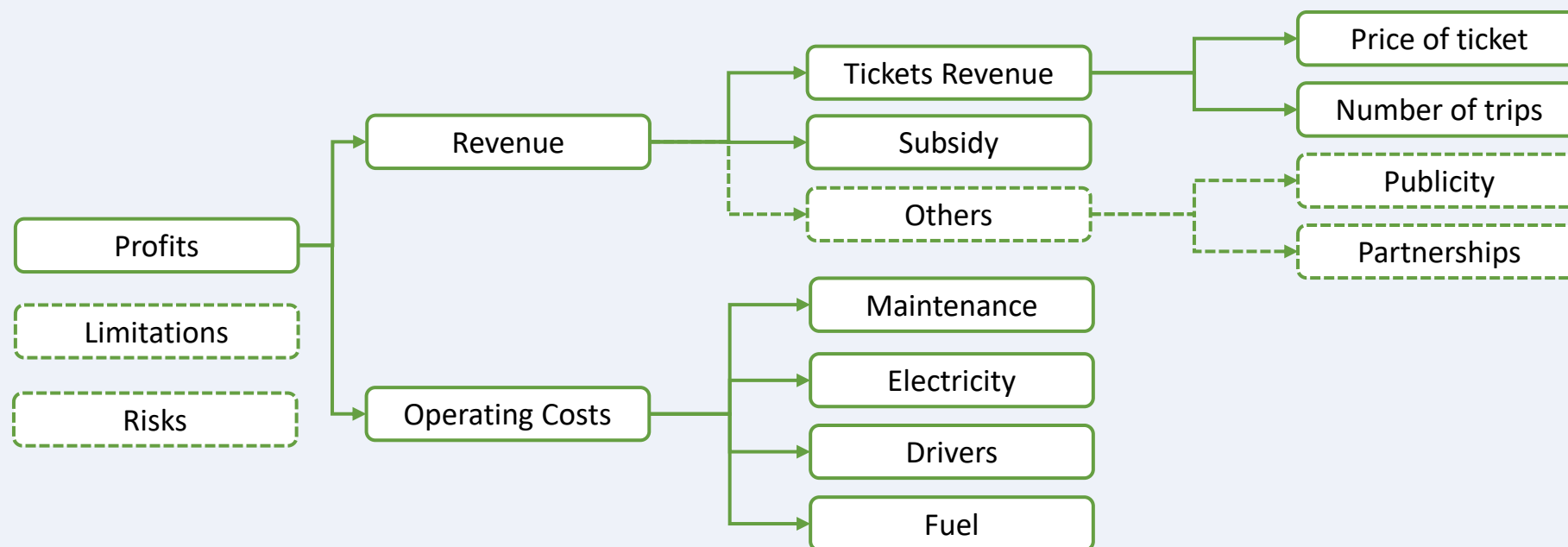
Since the case is talking about a service that has a big social impact, the interviewee is expected to be aware of the consequences of each of his decisions and try to pick the solution the impacts the less.

## STRUCTURE GUIDANCE

The structure for this case should be mainly about profit. The more related to transportation, the better. A good candidate should also consider all the risks and limitations involved in this process since we are talking about a case in the public sector under a lot of uncertainty. After the candidate presented the structure, handle Exhibit 1 or 2 depending on the candidate's request.

**Before handling Exhibit 2**, ask the candidate to do a **brainstorming of the main operating costs and possible covid-19 impact** on them.

## SAMPLE STRUCTURE



# Transantiago – Exhibit 1

Transportation ..... Easy  
Profitability ..... **Medium**  
Public Sector ..... Hard

## DEMAND FORECAST

|                               | Normal Ticket | Students    |
|-------------------------------|---------------|-------------|
| Total Trips projected 2020    | 500 million   | 100 million |
| Expected price                | 2 USD         | 1 USD*      |
| <b>Scenarios Covid Impact</b> |               |             |
| Optimist (30%)                | -10%          | -30%        |
| Neutral (40%)                 | -20%          | -60%        |
| Pessimist (30%)               | -30%          | -100%       |

*\*Note: Prices for Students can not be modified during 2020*

## DEMAND FORECAST ANALYSIS

From Exhibit 1 the candidate should understand the differences between each passenger and each one of the scenarios. A good candidate should be aware of the consequences of only considering an average scenario since there is a significant risk if the pessimistic scenario happens. Also, it is expected that the candidate takes some shortcut to calculate the revenue of each type of ticket, special for the “Normal Ticket”.

## NEW EXPECTED REVENUE

**Revenue Normal Tickets** = Normal Price \* Expected Demand = 2 USD \* 400 MM = **800 MM USD**

Expected Demand = Projected Demand \* (1+Expected Impact Covid) = 500 MM \* (1-20%) = 400 MM

Expected Impact Covid =  $0,3 \cdot -10\% + 0,4 \cdot -20\% + 0,3 \cdot -30\% = -20\%$

**Revenue Student Tickets** = Students Price \* Expected Demand = 1 USD \* 37 MM = **37 MM USD**

Expected Demand = Projected Demand \* (1+Expected Impact Covid) = 100 MM \* (1-63%) = 37 MM

Expected Impact Covid =  $0,3 \cdot -30\% + 0,4 \cdot -60\% + 0,3 \cdot -100\% = -63\%$

**TOTAL REVENUE** = Revenue Normal Tickets + Revenue Student Tickets = **837 MM USD**

# Transantiago – Exhibit 2

Transportation ..... Easy  
Profitability ..... **Medium**  
Public Sector ..... Hard

## INITIAL COSTS PROJECTION

| Total Costs*         |         | MM USD | 2,000        |
|----------------------|---------|--------|--------------|
| <b>Buses</b>         |         |        | <b>1,500</b> |
| Maintenance          |         |        | 300          |
| Fuel                 |         |        | 500          |
| Total Kilometres     | MM km   |        | 1000         |
| Consumption          | Lt/km   |        | 0,5          |
| Price of oil         | USD/lit |        | 1            |
| Bus drivers          |         |        | 500          |
| Other costs          |         |        | 200          |
| <b>Subway</b>        |         |        | <b>500</b>   |
| Maintenance          |         |        | 100          |
| Electricity          |         |        | 100          |
| Total Kilometres     | MM km   |        | 200          |
| Consumption          | Kwh/km  |        | 1            |
| Price of electricity | USD/Kwh |        | 0,5          |
| Subway drivers       |         |        | 200          |
| Other costs          |         |        | 100          |

\*Note: This cost where projected before Covid-19 effects

## Covid-19 Effects

- Average **price of oil** is expected to decrease 20% versus the initial projected price
- Average **price of electricity** is expected to decrease 10% versus the initial projected price
- Night curfews we will have two effects (**only for buses**):
  - Bus **driver's cost** will decrease 10% because fewer part-time drivers will be required
  - The total amount of **kilometres of buses** will be reduced by 10%

## COSTS ANALYSIS

**Brainstorm** - Before handling Exhibit 2, the candidate is asked to do a brainstorming of the main operating costs and possible covid-19 impact on them. For the brainstorm, the candidate is expected to mention most of the costs shown in the exhibit and some of the covid effects.

From Exhibit 2 the candidate is challenged to quickly understand all the information and see the relationship between each of the numbers with the covid-19 effect mentioned. Also, the candidate must understand the different types of units shown on the table.

## NEW EXPECTED COST

**New Fuel Cost** = New Total Kilometres \* Consumption \* New Oil Price = 900 MM \* 0.5 \* 0.8 = **360 MM USD**

New Total Kilometres = Original Total Kilometres \* (1 – Night Reduction) = 1000 MM \* (1-10%) = 900 MM

New Oil Price = Original Oil Price \*(1 - Oil Price Change) = 1 \* (1-20%) = 0.8

**New Electricity Cost** = Total Kilometres \* Consumption \* New Electricity Price = 200 MM \* 1 \* 0.45 = **90 MM USD**

New Electricity Price = Original Electricity Price \*(1 - Electricity Price Change) = 0.5 \* (1-10%) = 0.45

**New Bus Drivers Cost** = Original Bus Drivers Cost \* (1 – Night Reduction) = 500 MM \* (1-10%) = **450 MM USD**

**TOTAL COSTS** = Original Total Cost – Change in Fuel Cost – Change in Electricity Cost – Change in Bus Drivers Cost = 2000 MM – 140 MM – 10 MM – 50 MM = **1800 MM USD**

Change in Fuel Cost = Original Fuel Cost – New Fuel Cost = 500 MM – 360 MM = 140 MM USD

Change in Electricity Cost = Original Electricity Cost – New Electricity Cost = 100 MM – 90 MM = 10 MM USD

Change in Bus Drivers Cost = Original Bus Drivers Cost – New Bus Drivers Cost = 500 MM – 450 MM = 50 MM USD

## PROFIT ANALYSIS

After analysing all the revenue and costs, the candidate should be able to link both and **conclude that there will be a deficit of -63 MM USD**. The candidate is expected to start thinking about options to cover the expected deficit for this year. A good candidate will also mention that all these numbers are based on an average scenario and that the final difference could be covered or even increase depending on many of the variables.

**Ask the candidate to do a brainstorm of possible solutions and, if there is time, ask him to quantify one of them.**

A good candidate would deliver his **brainstorm in a structured way**.

## POSSIBLE SOLUTIONS

### Revenue solutions

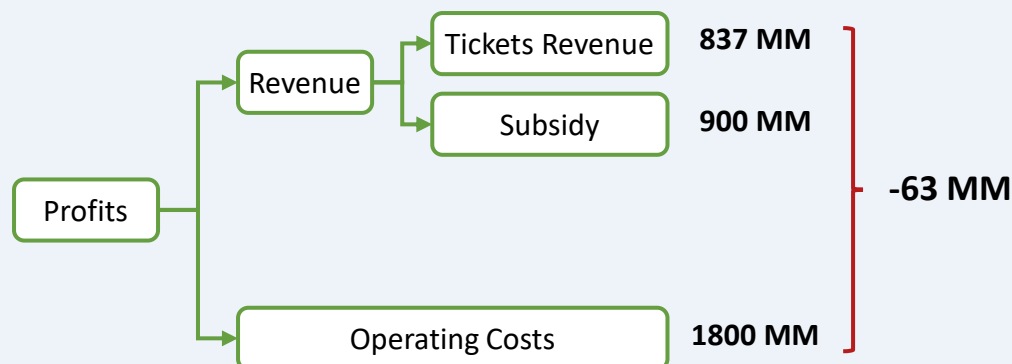
- **Increase the price of the “Normal Ticket”** – the price should be increased by 13 cents (6%)
- Increase safety measures to **generate more demand** – it would be required an increase of 31,5 MM “normal tickets”
- **Ask for more Subsidy** – An increase of 63 MM USD (7%)
- Create **other source of revenue**

### Cost solutions

- **Decrease the kilometres to reduce costs** – it would be required a decrease of 160 MM kilometres of Bus (17%)

### Other

- **Wait until July** to see which of the scenarios occur before taking actions that may not be necessary



## RECOMMENDATION QUESTION

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**The Ministry of Transportation has entered the room and is asking you for your analysis and recommendation.**

## SAMPLE RECOMMENDATION

---

The recommendation should be given in a structured way. Starting with the concrete action, then the reason behind that action, and after that, mention the risks involved. A possible recommendation structure would be:

- 1) Action/conclusion:** Mention that, after the analysis, we will have a gap of 63 million and bring up one of the possible solutions from the final brainstorm.
- 2) Reason:** Give two or three reasons why he thinks that is a good solution for the problem.
- 3) Risks:** Assess the possible negative impacts of the recommendation. A good candidate should also mention the risk behind all the numbers analysed, since they are all based on forecasts and could have high variability.
- 4) Recommendation:** Give a future recommendation to improve the analysis or to increase the action's probabilities of success.

# California Wildfires

---

*By Alan Bleiberg*



|                    |       |               |
|--------------------|-------|---------------|
| Public Sector      | ..... | Easy          |
| Climate Change     | ..... | <b>Medium</b> |
| Strategic Response | ..... | Hard          |

# California Wildfires

## PROMPT

The state of California has experienced devastating wildfires in recent years, causing deaths and billions of dollars worth of damage. The disasters are only projected to intensify, and the government needs to act fast. The governor has hired you to propose an action plan for prevention and mitigation. **What factors would you analyse?**

## STRUCTURE GUIDANCE

Given the open-ended nature and scale of the problem, there are a variety of approaches. However, the candidate's structure should cover the following key aspects of the problem. **Push the candidate to brainstorm and elaborate on ideas**, as the rest of the case is speculative.

### Diagnosing Crisis

Causes of Fire

Geographic Spread

Trends Over Time

- What are the **causes of wildfires** and **where** do they occur?
- Are certain causes **more preventable** than others?
- Are come types of wildfires **more devastating** than others?
- **Understand any trends** (time of year, certain geography, identifiable causes)?

### Action Measures

Prevention

Response

Time Horizon

- Understand what are **current prevention methods** and what are **addressable gaps**?
- What are the **current response tactics**? What new or existing tactics should be implemented?
- Action measures should be categorized by time-horizon: **short-term quick fixes** and **long-term infrastructure changes**
- What is the **effectiveness** of various strategies?

### Feasibility

Financial Cost

Ability to Implement

Legal and Regulatory

- What are the **costs** and **ability to implement** the proposed solutions?
- Are there **legal barriers** from stakeholders (lawmakers, utility companies, private sector)?
- Given scale, feasibility should be factor considered for each solution.

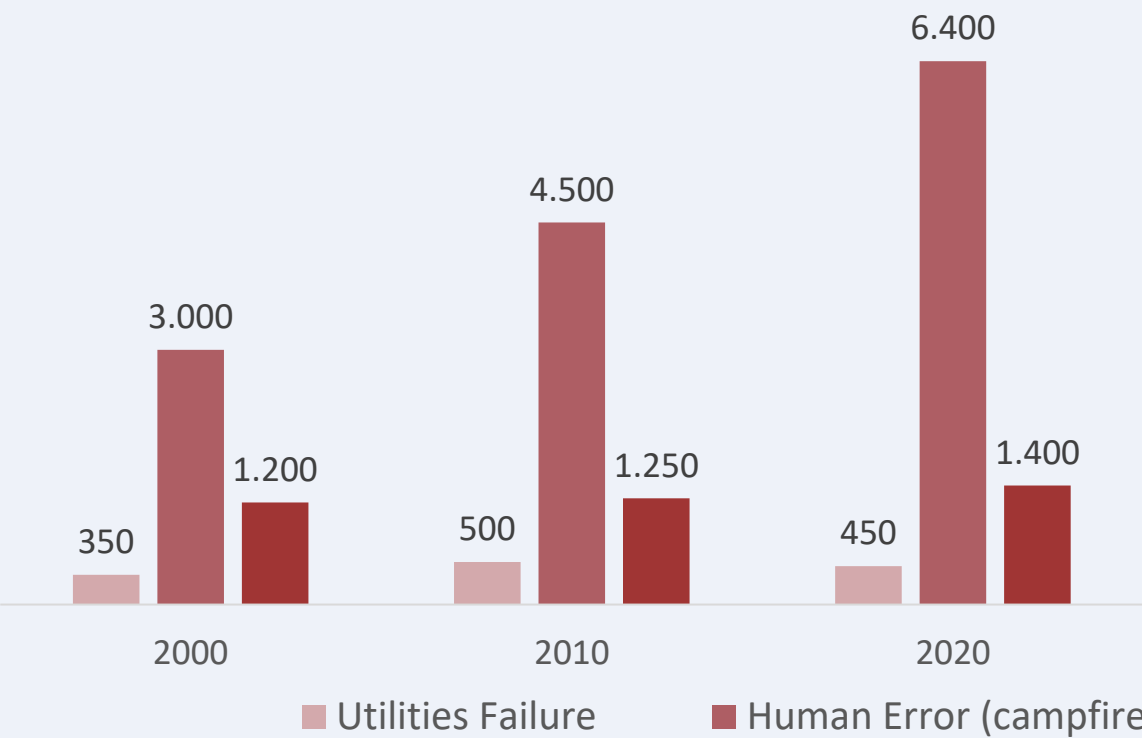
# California Wildfires – Exhibit 1

Public Sector  
Climate Change  
Strategic Response

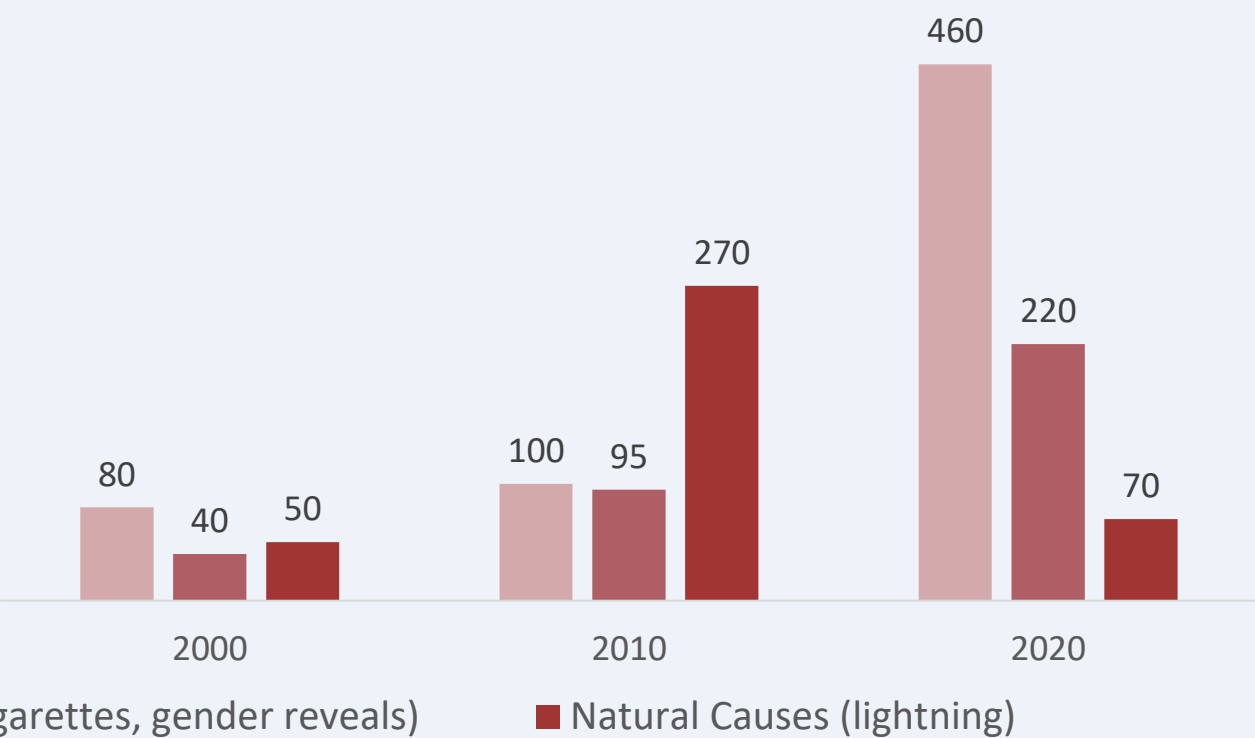
Easy  
Medium  
Hard

## CALIFORNIA HISTORICAL WILDFIRE REPORT

Number of Fires by Source



Total Area Burned by Source (Thousand Acres)



# California Wildfires – Exhibit 1

## INSIGHT ANALYSIS

Hand over Exhibit 1 if the interviewee asks about **causes of wildfires or trends over time**

## EXPECTED TAKEWAYS FROM EXHIBIT

**Candidate should note the temporal data set and identify trends and outliers**

- Candidate should explain that **fires due to human error have increased** over the last 20 years (2x), while other causes of fires are relatively constant
- Next, candidate should note two observations on the quantity of acres burned:
  1. In-line with increased quantities of fires, **acres burned due to human error has steadily increased** over the last 20 years
    - a) Candidate should note these are **likely frequent but smaller fires**, hence the relative magnitude is not extremely high
    - b) Candidate should highlight this is an area to focus on prevention, as the cause is avoidable and various policies should be able to reverse the increase
  2. The candidate should also note outlier years, **2010 (natural causes) and 2020 (utilities)**
    - a) Candidate should speculate these were likely **single “disaster” fires** that inflate the year’s statistics
    - b) Candidate should think these are likely impossible to prevent in the short-term due to the nature of the cause, but could suggest long-term prevention or mitigation measures
- Let the candidate think out loud and drive ideas, but next **present Exhibit 2**

# California Wildfires – Exhibit 2

## 2022 CALIFORNIA DEPARTMENT OF FORESTRY ANALYSIS

| City          | Cost to Build SafeWall <sup>1</sup> (\$M) | Residential Structures | % of Structures At-Risk | Annual Wildfires | % that are catastrophic <sup>2</sup> | Avg Property Value (\$M) |
|---------------|-------------------------------------------|------------------------|-------------------------|------------------|--------------------------------------|--------------------------|
| Oakland       | 6.4                                       | 400,000                | 2%                      | 10               | 5%                                   | 1.00                     |
| Santa Barbara | 7.0                                       | 100,000                | 4%                      | 15               | 15%                                  | 2.50                     |
| Sacramento    | 5.8                                       | 250,000                | 3%                      | 12               | 10%                                  | 0.50                     |

### Notes

1. SafeWall reduces the probability of a catastrophic fire by 50%
2. “Catastrophic” defined as destroying 10% of at-risk structures

# California Wildfires – Exhibit 2

## QUANTITATIVE ANALYSIS

An environmental minister has suggested that a fireproof structure, SafeWall, could be built and would reduce the probability of a catastrophic wildfire. To start, they can only build around one city – which should they build and why?

## SOLUTION

**Candidate should assess each option and systematically compare the expected outcomes**

**Step 1:** Calculate the **at-risk value** in each location:

*Residential Structures \* % At Risk \* Avg Property Value*

#1

| City          | Residential Structures | % of Structures At-Risk | Total At-Risk Structures | Avg Property Value (\$M) | Total At-Risk Value (\$B) |
|---------------|------------------------|-------------------------|--------------------------|--------------------------|---------------------------|
| Oakland       | 400,000                | 2%                      | 8,000                    | 1.00                     | 8.00                      |
| Santa Barbara | 100,000                | 4%                      | 4,000                    | 2.50                     | 10.00                     |
| Sacramento    | 250,000                | 3%                      | 7,500                    | 0.50                     | 3.75                      |

**Step 2:** Calculate the **likelihood of catastrophic fire** in each location

*Annual Wildfires \* % that are catastrophic*

#2

| City          | Annual Wildfires | % that are catastrophic | Expected Catastrophic Fires |
|---------------|------------------|-------------------------|-----------------------------|
| Oakland       | 10               | 5%                      | 0.5                         |
| Santa Barbara | 15               | 15%                     | 2.25                        |
| Sacramento    | 12               | 10%                     | 1.2                         |

**Step 3** Multiply the **probability of a catastrophic fire** by the **at-risk value** in each location by **10% destruction. Reduce expected damage by 50% to determine Value Saved**

*At Risk Value \* Expected Wildfires \* 10% Destruction; → \* 50% reduction and Compare Values*

**Expected Insight:**

- Candidate should note that **SafeWall in Santa Barbara** would save the most value (**\$1B**) compared to the other locations (~5x)
- Candidate **should take a stance** on basing the decision on financial considerations compared to population or at-risk structures. Candidate should note more data is needed on population.
- Should note that construction costs are not a factor given relatively similar in all locations

#3

| City          | Total At-Risk Value (\$B) | Expected Catastrophic Fires | Damage of Catastrophic Fire | Expected Cost, No SafeWall (\$M) | Impact of SafeWall | Expected Cost, SafeWall (\$M) | Value Saved (\$M) |
|---------------|---------------------------|-----------------------------|-----------------------------|----------------------------------|--------------------|-------------------------------|-------------------|
| Oakland       | 8.00                      | 0.5                         | 10%                         | 400                              | 50%                | 200                           | 200               |
| Santa Barbara | 10.00                     | 2.25                        | 10%                         | 2,250                            | 50%                | 1,125                         | 1,125             |
| Sacramento    | 3.75                      | 1.2                         | 10%                         | 450                              | 50%                | 225                           | 225               |

# California Wildfires

## RECOMMENDATION QUESTION

Between your structured approach & data provided, what are your recommendations for the governor?

## SAMPLE RECOMMENDATION

Allow for flexibility based on the candidate's structure. A **balance of creativity and practicality** should be embraced

- Based on data trends, action should be taken to **reduce the frequency of wildfires from human error**
- **Long-term solutions** should also be implemented to **mitigate “disaster” fires** that cause more damage
- Building SafeWall around **Santa Barbara** is expected to **save over \$1B** in real estate value, 5x the value of other locations
- Overall **response risks are plentiful**, namely effectiveness, cost, and ability to implement
- **Action steps should be relevant** to the prior conclusions and **include other proposed solutions**, such as:

| Short-Term Actions                                                                                                                                                                                 | Long-Term Changes (Infrastructure)                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Station more firefighters</li> <li>• Purchase more water-planes</li> <li>• Ban the use of campfires</li> <li>• More frequent forest management</li> </ul> | <ul style="list-style-type: none"> <li>• Update high-risk power lines</li> <li>• Clear break-lines in forests</li> <li>• Implement drone detection fleet</li> </ul> |

# South bank

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By *Giancarlo Young*



|                    |       |               |
|--------------------|-------|---------------|
| Financial Services | ..... | Easy          |
| Product launch     | ..... | <b>Medium</b> |
| Profitability      | ..... | Hard          |

## PROMPT

---

South Bank (SB) is the leading retail bank in Peru, an emerging South American country. SB has a dominant performance on high-income and a very good performance in medium-income customers but has not been capable of entering to low-income market. CMO has told us that his team has been evaluating the option to launch a new credit card with cashback benefits, which are perceived as much more valuable in this segment, and the product has already shown some results in competitors.

Currently, SB has a credit card with a loyalty program based on airline miles. This program is considered part of a strategic alliance with an important regional airline.

**The CMO would like you to evaluate if it is convenient to invest in this project.**

## CLARIFYING POINTS (if asked)

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- Peru has 35 M inhabitants. The local currency is PEN. Adults represent 70% of population.
- Country has 3 main economic segments, with clear different behaviours in credit card use.
- Consider Credit Card business as an independent unit of analysis.
- Revenues are only generated by merchant fee, a percentage of the amount paid with the card. During the last years has been stable at 2%.
- Each client can only have 1 credit card at the same time.
- To be approved, a project is required to have a payback period of 3 years. As a secondary metric, CMO prefers to generate the highest possible net cash flow in the first 3 years.

## CASE GUIDANCE

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The case requires the candidate to financially estimate (using payback period) the feasibility of launching this new product and assess the option of a new sales channel. After calculating both results, as a counterbalance, there are qualitative, country-related situations that have to be considered and open a chance to brainstorm, and even propose a different recommendation.

The candidate will need to ask for information to create a market sizing logic. Then, the interviewer can start showing exhibits. Is expected to work with order to avoid reprocessing and don't get confused with numbers.

## STRUCTURE GUIDANCE

- Candidate should focus on main aspects that imply to release a new product: financial impact, market situation, risks involved, and capabilities that the company has.

## SAMPLE STRUCTURE

### FINANCIALS

- I) Revenues: # customers, frequency of use, size of ticket, merchant fee value.
- II) Costs: variable -> sales commissions, plastic cards, loyalty program; fixed -> wages, system maintenance

### MARKET

- I) Competitors: BS market share, positioning, value proposition
- II) Clients: segments, purchase behaviours and perceptions
- III) Ecosystem: availability of POS, trends in cashless options

### CAPABILITIES

- I) Commercial: brand awareness, know how of regional needs and customs
- II) Financial: Budget for investment, need of additional CAPEX/OPEX
- III) Operational: scale and granularity of sales force, logistics for credit card supply

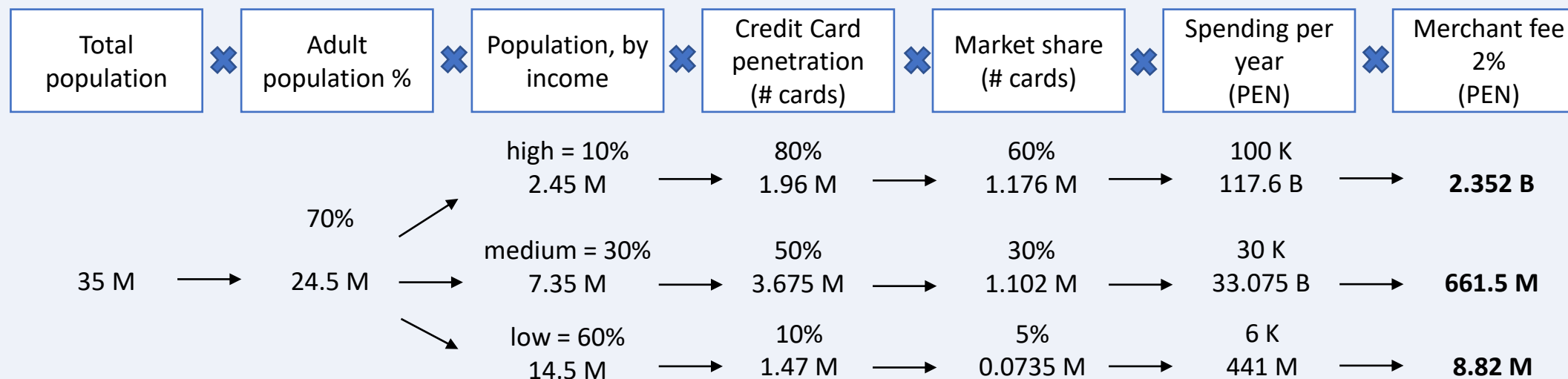
### RISKS

- I) Financial: small ROI
- II) Commercial: bad experience of user because of lack of knowledge regarding credit cards or not enough POS to pay, cannibalization vs current credit card

## QUANTITATIVE ANALYSIS, PART 1

- Candidate should propose a market sizing as first action, explaining the reasoning before doing numbers. Interviewer should show EXHIBIT 1 once the candidate has identified many variables implied. Candidate should mention macro features (e.g. large market vs small ticket and small penetration)
- After having calculated revenue per segment, challenge the idea (*so what?*) of entering low income since is a very small market (3% of total, 0.3% of BS revenues).
- If the candidate only considers calculating values for low-income segment, is adequate. Still, he/she should calculate all segments in order to compare them and find some insights of the whole potential.

## SOLUTION, PART 1



# South bank – Exhibit 1

---

Financial Services ..... Easy  
Product launch ..... **Medium**  
Profitability ..... Hard

## CREDIT CARD MARKET MAIN INDICATORS

---

| Segments      | % of population | Credit card penetration | SB market share | Avg. spending per year |
|---------------|-----------------|-------------------------|-----------------|------------------------|
| High-income   | 10%             | 80%                     | 60%             | 100,000                |
| Medium-income | 30%             | 50%                     | 30%             | 30,000                 |
| Low-income    | 60%             | 10%                     | 5%              | 6,000                  |

## QUANTITATIVE ANALYSIS, PART 2

- Once the interviewee calculates the revenues, the interviewer has to indicate that we have receive additional information about 2 options of sales channels: traditional (branches) or online. Criteria for choosing an alternative remain the same as indicated during clarifying questions.
- Interviewer mentions main revenues and costs (see SOLUTION, PART 2).
- Once cash flows have been calculated, show (in EXHIBIT 2) that we don't have branches in the majority of low-income regions.

## SOLUTION, PART 2

- Compare marginal impact on branches vs online sales:
  - Market share growth (# cards) -> branches: from 5% to 15%; online: from 5% to 8%. Average spending and merchant fee remains the same.
  - Variable costs (paid only at year 0) -> plastic card emission = PEN 5 // sales commission (per card sold) = PEN 50
  - Variable costs (paid every period) -> cashback = 1% of payments, per year
  - Fixed costs (paid every period) -> online channel maintenance = PEN 200 K per year
  - **Net cash flow after 3 years (PEN) -> Branches = 18.4 M vs. Online = 7.2 M. Payback time = 1 year for both options.**

| Branches // # new cards = 147 K | 0         | 1-3      |
|---------------------------------|-----------|----------|
| Payments                        |           | 882 M    |
| Revenue                         |           | 17.64 M  |
| Costs (plastic & sales)         | (8.085 M) | 0        |
| Cashback                        |           | (8.82 M) |
| Margin                          | (8.085 M) | 8.82 M   |

| Online sales // # new cards = 44.1 K | 0         | 1-3       |
|--------------------------------------|-----------|-----------|
| Payments                             |           | 264.6 M   |
| Revenue                              |           | 5.29 M    |
| Costs (plastic & online)             | (0.735 M) | (0.200 M) |
| Cashback                             |           | (2.65 M)  |
| Margin                               | (0.735 M) | 2.65 M    |

# South bank – Exhibit 2

|                    |       |               |
|--------------------|-------|---------------|
| Financial Services | ..... | Easy          |
| Product launch     | ..... | <b>Medium</b> |
| Profitability      | ..... | Hard          |

## PERU'S POLITICAL MAP



- Lima City concentrates around 10M inhabitants (disregard the segment), medium-income regions represent other 10M inhabitants, and 15M inhabitants live in low-income regions.
- Branch penetration is usually preceded by a developed POS ecosystem, implemented by a Credit Card global company.
- Technologies adopted by low-income segment: 65% has a smartphone, 65% has Internet access.

- ★ High penetration of SB branches
- High income regions
- Medium income regions
- Low income regions

# South bank – Exhibit 2

---

|                    |       |               |
|--------------------|-------|---------------|
| Financial Services | ..... | Easy          |
| Product launch     | ..... | <b>Medium</b> |
| Profitability      | ..... | Hard          |

## TAKEAWAYS

---

- The candidate should recognize that we don't have any presence in the majority of low-income regions. Thus, it would imply an additional investment in building branches.
- A good candidate should also consider the cash flow differences between sales channels ( $\text{PEN } 18.4\text{M} - 7.2\text{M} = 11.2\text{M}$ ) as additional CAPEX that could be used for building branches and still generating higher cash flows vs online.
- Since Lima City has the third part of Peru's population, is probable that there is a hidden low-income segment that we can prioritize. The same for other high-income, medium-income regions.
- Leverage on technologies could help to foster financial literacy or as payment methods.

## BRAINSTORMING QUESTION

---

1. What are the main potential problems/risks that we have in order to launch the new product?
2. What would you suggest to do to hedge these risks?

## SAMPLE ANSWERS

---

1. Candidate should bundle possible risks:
  - Commercial: acquiring the card through online sales could be daunting, clients don't understand how to use the card, 1% cashback could not be perceived as attractive, erosion of partnership with airline.
  - Financial: clients show high default levels, high-income clients turn to cashback credit card (cannibalization).
  - Operational: there are not enough POS nor branches in the majority of low-income regions, low density of population in low-income regions.
2. Bundling is, again, a good practice:
  - Commercial: partner with local authorities or leaders to give financial education and training, focus on capture competitors' clients using higher cashback or other discounts, revisit analysis and focus on high-density, higher-income regions.
  - Financial: evaluate if opening mini-branches (less CAPEX).
  - Operational: deliver credit cards by courier, negotiate with Credit Card provider/competitors/government to accelerate POS instalment, evaluate other technologies as Digital Credit Card (in smartphones).

## RECOMMENDATION QUESTION

---

**What is your recommendation to South Bank CMO?**

## SAMPLE RECOMMENDATION

---

- The candidate should recommend implementing the project. Payback period is **only 1 year regardless of sales channel**. Ideally, he/she should suggest implementing branches, because they generate higher net cash flows in the first 3 years.
- Consider that, if we have to invest **more than PEN 11.2 M** in opening new branches, is better to go online.
- As possible risks to consider: low adoption of product (due to access, lack of knowledge, or small POS network), potential cannibalization, and erosion on airline partnership.
- Finally, mention next steps: consider push POS penetration with Credit Card provider or even with competitors, work on financial education, etc.

### **A good candidate will:**

- Recognize that there is an underlying, structural problem that has to be solved to increase probability of success. He/she explains that the project is a long-run bet, since these markets have to be developed.
- Consider other costs/losses as defaults or frauds, and suggest a sensitivity analysis (e.g. % of market captured, payback period required = 1 year, higher cashback %).
- Size how many low-income clients are in medium or high-income regions. For instance, Lima City probably has a relevant low-income population, and already counts with a high branch penetration.
- Ponder to explore medium-income segment, since the size, penetration and market share put SB in a better position.

# CONTENT

- ❖ ACKNOWLEDGEMENT
- ❖ HOW TO USE THE CASE BOOK
- ❖ CASES 2<sup>nd</sup> Edition

- ❖ **CASES 1<sup>st</sup> Edition**



Cowbon Emissions *By Emily Hinton (IESE MBA 2021)*



African Gold Mines Co. *By Marc-Olivier Granger (IESE MBA 2021)*



Nica Productions *By Alfonso Tomás Durandeu (IESE MBA 2021)*



Pipeline Oil Technology *By Roberto Carlos De Araujo (IESE MBA 2021)*



The Bookstore *By Víctor Manzanares Bonilla (IESE MBA 2021)*



Gymco *By Pieter Swart (IESE MBA 2021)*



Green Airlines *By Antonio Niemeyer (IESE MBA 2021)*

# Cowbon Emissions

---

*By Emily Hinton (IESE MBA 2021)*



Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## PROMPT

Our client is a major milk producer in New Cowland, Milking it Co., MIC. New Cowland has recently introduced a law that means MIC has to reduce its GHG emissions by 45% of 2019 levels within the next five years or face being shut down or heavily fined. They currently produce 20% of New Cowland's GHG emissions.

**The CSO has hired us to figure out a way to reach this target.**

## CLARIFYING POINTS (if asked)

- MIC produces approximately 100% of New Cowland's milk supply
- MIC does not have plans for expansion, but reducing volumes is not an option
- They only produce milk and have no plans to diversify
- Budget for this project is \$750m/year for the next five years (for perspective, current revenues are \$15billion)
- MIC owns the entire production chain – from farms, production and transport, they sell to a variety of clients

*The candidate might ask what the breakdown of where GHG comes from within the business, this is shown in Exhibit 1*

- 5% of their market is local, the rest is foreign
- Only need to reduce GHG directly produced by MIC

## CASE GUIDANCE

This case is designed to test brainstorming, business decision-making skills and logic. It will help candidates wanting to practice market sizing and working on unconventional problems. For calculations ignore the time value of money.

It is a long case designed for advanced candidates; some aspects can be removed for the sake of time – these are clearly marked.

Interviewer guidance has been provided at various stages.

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## INTERVIEWER GUIDANCE - STRUCTURE

**A good framework would touch on:**

- Different ideas to reduce GHG emissions in the different segments of the business (farming, processing, transportation, overheads, etc.)
- Ability to implement changes
- Financial implications
- Non-financial implications – PR, risks (e.g., change in government, a backlash from farmers)

*The candidate should lead the case towards understanding the current breakdown of GHG emissions – if not, the interviewer should gently nudge toward this path.*

## GHG EMISSIONS ANALYSIS

**The candidate should be presented with Exhibit 1 if s/he asks about the breakdown of GHG in each business segment.** Candidate should be told that MIC has already taken measures to reduce Overheads Emissions and MIC believes that there is no further that can be done to reduce these.

## EXHIBIT 1 TAKEAWAYS

- GHG emissions have been increasing over the last three years with farming being the key driver
- Farming is the largest emitter followed by production
- From farming, cows are by far the largest emitter producing 63% of 2019's emissions – this would be the key thing to look into first as it can make the biggest difference (it is the only value that is over 50% of the emissions by itself)
- Processing is the second biggest emitter so that should be focused on next (20% of total emissions)

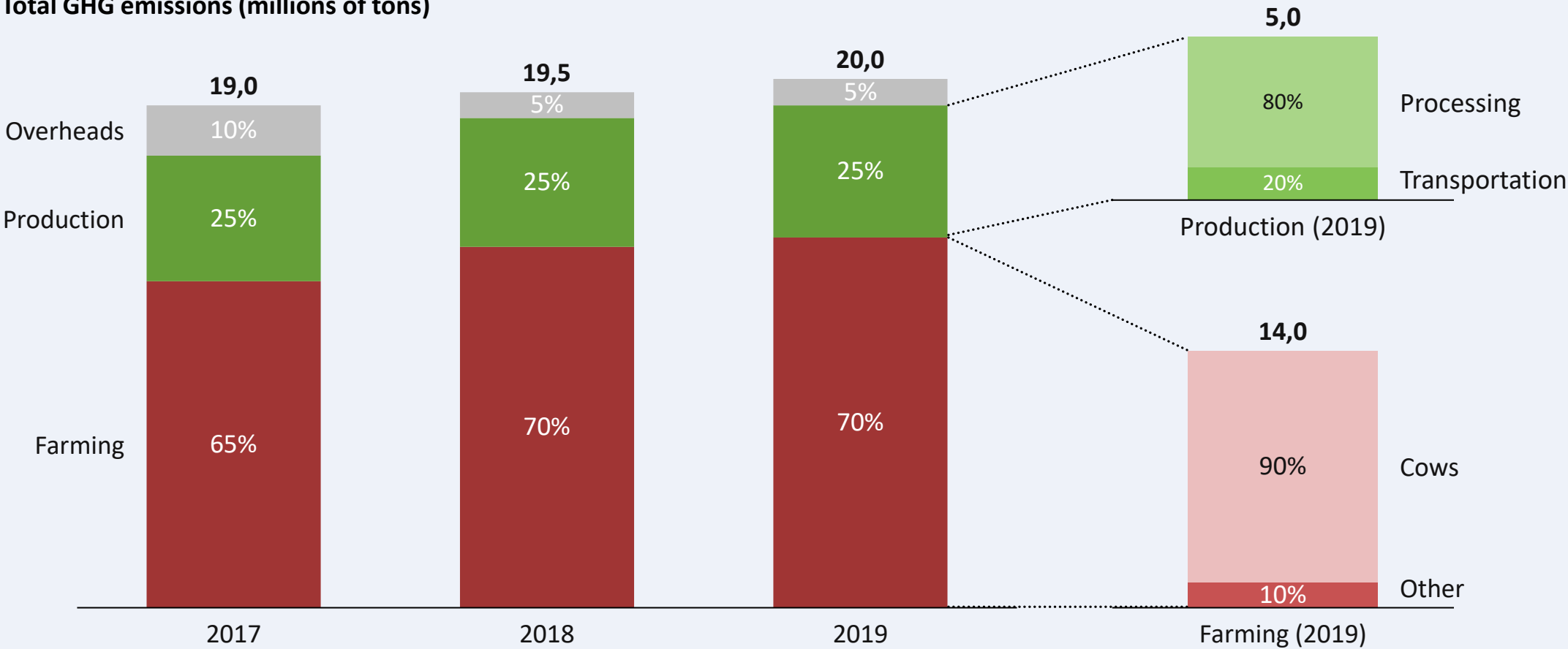
*The candidate should identify cows as the first logical step to explore in reducing emissions.*

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... Hard

## EXHIBIT 1 - TRENDS OF GHG EMISSIONS OF MIC

Total GHG emissions (millions of tons)



# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## COW EMISSION ANALYSIS

There is a new breed of cow that MIC is interested in to replace their current herd. All characteristics are the same as the current herd, but they produce 33% less GHG emissions.

**As MIC owns farms all over New Cowland, they want us to approximate how many cows they have in total.**

## SUGGESTED CALCULATION

**\*\* if the interview is progressing slowly, skip this estimation and give the number of cows of 5 million, then move straight to part 2 after reading the prompt below\*\***

This is an estimation problem – the candidate should recognize the need to size the herd using suitable assumptions, provide the estimation data only if the candidate asks.

## ESTIMATION DATA

### **Provide only if requested:**

- Population of New Cowland: 5 million
- Milk produced per cow: 10L/days
- Number of days in a year: 300
- Domestic market: 5% of sales
- Domestic market share: 100%
- Average milk consumption per person: 150L/year

### **Total annual domestic consumption:**

$[\text{Milk consumption per person}] \times [\text{population of NZ}]$   
= 750 million L/year

### **Cows required for domestic consumption:**

$[\text{Annual dom. Cons.}] / ([\text{Daily pro./cow}] \times [\text{days in a year}])$   
= 250,000 cows

### **Total cows:**

$[\text{domestics cows}] / [\% \text{ of market}]$   
= **5 million cows**

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## COW EMISSION ANALYSIS

Knowing that there are 5 million cows in the MIC herd and the average life of a dairy cow is 5 years. **Calculate the total cost of replacing the herd and the total emissions saved from this replacement.**

- The old breed cost \$2000 and the new breed costs \$2500 to purchase
- The old breed emits 2.5 ton/year and the new breed emits 33% less GHG

## SUGGESTED CALCULATION & TAKEAWAYS

|                   | YEAR 1                                                                                                  | YEAR 2                                                                                                 | YEAR 3                                                                                                  | YEAR 4                                                                                                    | YEAR 5                                                                                                  |
|-------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Cost              | [1 million cows]<br>*[\$500/cow]<br>= \$500 million                                                     | [1 million cows]<br>*[\$500/cow]<br>= \$500 million                                                    | [1 million cows]<br>*[\$500/cow]<br>= \$500 million                                                     | [1 million cows]<br>*[\$500/cow]<br>= \$500 million                                                       | [1 million cows]<br>*[\$500/cow]<br>= \$500 million                                                     |
| GHG ton reduction | [1 million cows]*<br>[0.33*2.5t]<br><u>Or</u><br>[12.6million tons]*<br>[33%]/[5/1]<br>= 0.825 mil tons | [2 million cows]*<br>[0.33*2.5t]<br><u>Or</u><br>[12.6million tons]*<br>[33%]/[5/2]<br>= 1.65 mil tons | [3 million cows]*<br>[0.33*2.5t]<br><u>Or</u><br>[12.6million tons]*<br>[33%]/[5/3]<br>= 2.475 mil tons | [4 million cows]*<br>[0.33*2.5t]<br><u>Or</u><br>[12.6million tons]*<br>[33%]/[5/4]<br>= 3.3 million tons | [5 million cows]*<br>[0.33*2.5t]<br><u>Or</u><br>[12.6million tons]*<br>[33%]/[5/5]<br>= 4.125 mil tons |
| % GHG reduction   | 4.125%                                                                                                  | 8.25%                                                                                                  | 12.375%                                                                                                 | 16.5%                                                                                                     | 20.625% (~21%)                                                                                          |

Candidate can calculate only Year 5 as this is the final number required but must check that the annual budget is not exceeded.

**Candidate should conclude that this is approximately halfway to our goal, spending 67% of our budget.**

**Next should look at reducing emissions from production.**

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## PRODUCTION EMISSIONS ANALYSIS

What are some ways that the GHG emissions can be reduced during the production phase?

## BRAINSTROMING SAMPLE

**\*\* if the interview is progressing slowly, skip this brainstorming and move straight to part 2 after reading prompt 2 on this page\*\***

This brainstorm is a chance for the candidate to be creative. No structure is superior but looking at options split between processes and transportation is one way to go. Keep pushing until you are satisfied with the ideas generated.

| PROCESSES                                 | TRANSPORTATION                              |
|-------------------------------------------|---------------------------------------------|
| Alternative power sources (all renewable) | Larger trucks (i.e. less GHG/L transported) |
| Carbon capture of emissions               | Electric vehicles                           |
| More energy-efficient processes           | Shorter routes of transportation            |
| R&D into different ways to process milk   | Regular maintenance to increase efficiency  |
| Outsource production                      | Outsource transportation                    |
| Shutdown inefficient plants               |                                             |

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... Hard

## PRODUCTION EMISSIONS ANALYSIS

MIC has been looking into different ways to reduce GHG emissions in both processes and transportation and they have identified these options (show Exhibit 2).

## CALCULATION AND TAKEAWAYS – EXHIBIT 2

This exhibit is intentionally challenging to understand, point the candidate towards the footnotes if they are struggling to understand the meaning of each column.

### Key insights from Exhibit 2:

- Conversion to renewable energy has the largest emissions savings (potential reduction of 14% of GHG) and can be completely replaced within the 5-year period
- Replacing sterilizing units will only reduce emissions by 4% total and will take 20 years to carry out
- The other processes are also not cost-effective and should be written off straight away
- Transportation is a cost-effective method of GHG reduction which can be completed within 5 years and will reduce emissions by 5% total

|                           | COST PER YEAR | TOTAL COST (5 YEARS)                | GHG REDUCTION PER YEAR                        | TOTAL GHG REDUCTION (5 YEARS)            |
|---------------------------|---------------|-------------------------------------|-----------------------------------------------|------------------------------------------|
| Renewable energy source   | \$150 million | [\$150m] * [5years] = \$750 million | [25%]*[80%]*[70%]*[20%]= 2.8% (0.56 mil tons) | [2.8%] * [5 years] = 14% (2.8 mil tons)  |
| New Sterilizing Process   | \$100 million | [\$100m] * [5years] = \$500 million | [25%]*[80%]*[20%]*[5%]= 0.2% (0.04 mil tons)  | [0.2%] * [5 years] = 0.8% (0.2 mil tons) |
| Transport electrification | \$50 million  | [\$50m] * [5years] = \$250 million  | [25%]*[20%]*[100%]*[20%]= 1% (0.2 mil tons)   | [2.8%] * [5 years] = 5% (1 mil tons)     |

The candidate should conclude that conversion to renewable energy and electrifying the transport fleet will be the most efficient costing \$200m/year and reducing the GHG emissions by 19% in total by year 5

# Cowbon Emissions

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... Hard

## EXHIBIT 2 - ALTERNATIVES TO REDUCE GHG EMISSIONS OF MIC

|                  | PROPOSED<br>CHANGE TO<br>PRODUCTION | TOTAL %<br>REDUCTION OF<br>EMISSIONS <sup>1</sup> | # OF UNITS <sup>2</sup> | MAX. % ANNUAL UNIT<br>REPLACEMENT <sup>3</sup> | ANNUAL TOTAL COST<br>OF MAX REPLACEMENT <sup>4</sup> |
|------------------|-------------------------------------|---------------------------------------------------|-------------------------|------------------------------------------------|------------------------------------------------------|
| <b>Process</b>   | Renewable energy source             | 70%                                               | 10                      | 20%                                            | \$150m                                               |
|                  | New Sterilizing Process             | 20%                                               | 20                      | 5%                                             | \$100m                                               |
|                  | New Packaging process + material    | 2%                                                | 50                      | 20%                                            | \$50m                                                |
|                  | Other                               | 0.1%                                              | 500                     | 10%                                            | \$1m                                                 |
| <b>Transport</b> | Electrification of fleet            | 100%                                              | 500                     | 20%                                            | \$50m                                                |

1 % of reduction in GHG emissions that the process or transportation will emit, based on the current emissions shown in Exhibit 1, once 100% of the units have been replaced

2 number of units MIC currently owns that can be replaced in a 1:1 ratio with the new alternatives

3 % of units that can be replaced annually (due to end of life requirements) e.g. a total of 2 units can be replaced per year for 'renewable energy source'

4 How much it will cost MIC annually to replace the maximum allowable units e.g. will cost MIC \$150m to replace 2 units of 'renewable energy source' per year

# Cowbon Emissions

---

Agriculture ..... Easy  
Sustainability ..... Medium  
Operations ..... **Hard**

## RECOMMENDATION

---

**Great, the CSO is about to join us, can you please provide her with a brief summary of what we have discussed today?**

## SAMPLE RECOMMENDATION

---

**The candidate should summarize that:**

- We should replace the herd with the new breed to reduce emissions by 21% in year 5, costing \$2.5 billion
- We should convert to renewable energy and replace our transport fleet reducing our emissions by 19% in year 5, costing \$1 billion
- These actions will get us 90% towards our target of 45% reduction
- We have \$50m/year of our budget left to figure out the last 10% which could include.... (any ideas that you have discussed e.g., carbon credits, hiring a lawyer to reduce potential fines, other emission reductions)

**A great candidate would also briefly discuss any risks or next steps:**

- Adaption to new technology
- Chance new breed has challenges
- Views from employees and farmers about changes
- Chance of lobbying government so the law is reversed
- Next steps: contact breeders to make sure they have enough cows, look for other reduction methods, etc.

# African Gold Mines Co.

---

*By Marc-Olivier Granger (IESE MBA 2021)*



|               |             |
|---------------|-------------|
| Mining        | Easy        |
| Operations    | Medium      |
| Profitability | <b>Hard</b> |

# African Gold Mines Co.

Mining ..... Easy  
Operations ..... Medium  
Profitability ..... **Hard**

## PROMPT

African Gold Mines Co. (AGM) is a junior gold producer with one open-pit mine currently in production in Cameroon. The mine was established 10 years ago, and forecasts predict at least another 10 years of consistent gold production. Last year, AGM produced 250k oz of gold from their mine. They have come to us as their profit level has been slowly decreasing over the past five years. They had numerous operational issues with the mining equipment, problems with the workers' union and issues with community relations notably.

**They want us to investigate what is the best course of action for their operation to return to their previous level of profitability within 12 to 24 months.** They want us to specifically focus our attention to the mining department of the company.

## CLARIFYING POINTS (if asked)

- Costs have increased by 20% over the past 5 years
- Mining department costs in 2019 were \$36m
- Revenues for the entire company in 2019 were \$375m
- Target is to return to profitability level of 5 years ago meaning reducing costs by 6m.
  - Cost 5 years ago =  $36/1.2=30$  Cost reduction =  $36-30=6m$

## CASE GUIDANCE

This is an interviewee-led case and the candidate should always drive the case and suggest the next course of action.

This is a profitability case that focuses on the cost side. Some information can be given first regarding revenues, but no conclusion can be drawn.

The candidate should have some form of a profitability tree as a structure. The more specific to mining, the better.

**Once the structure is done, direct the interviewee to look into revenues first, then, hand over Exhibit 1.**

Looking at costs, the candidate should identify that the best opportunity to reduce cost is to examine equipment maintenance & fuel consumption.

From there, the candidate will be informed that AGM has been looking at changing its fleet of trucks and be given the four alternatives to choose from.

# African Gold Mines Co.

---

Mining ..... Easy  
Operations ..... Medium  
Profitability ..... Hard

## REVENUE ANALYSIS

---

Regarding revenues, our client has done some analysis. **Here is a chart of AGM's historical gold production and yearly average gold price (Show Exhibit 1).**

## TAKEAWAYS – EXHIBIT 1

---

From Exhibit 1, the interviewee should understand that despite the variation in price and production the total revenues have remained about the same. Furthermore, gold prices, like for any other commodities, are decided on the markets and there is very little that AGM can do to influence the prices. In addition, geology dictates the volume of gold that can be mined each year, this is part of the long-term plan of the company and cannot be altered. From this conclusion, the candidate should push the case into the cost branch of the tree.

## REVENUE ANALYSIS

---

**What do you think are the biggest costs drivers as part of such mining operations? What do you think we should focus our attention on?**

## BRAINSTORMING SAMPLE

---

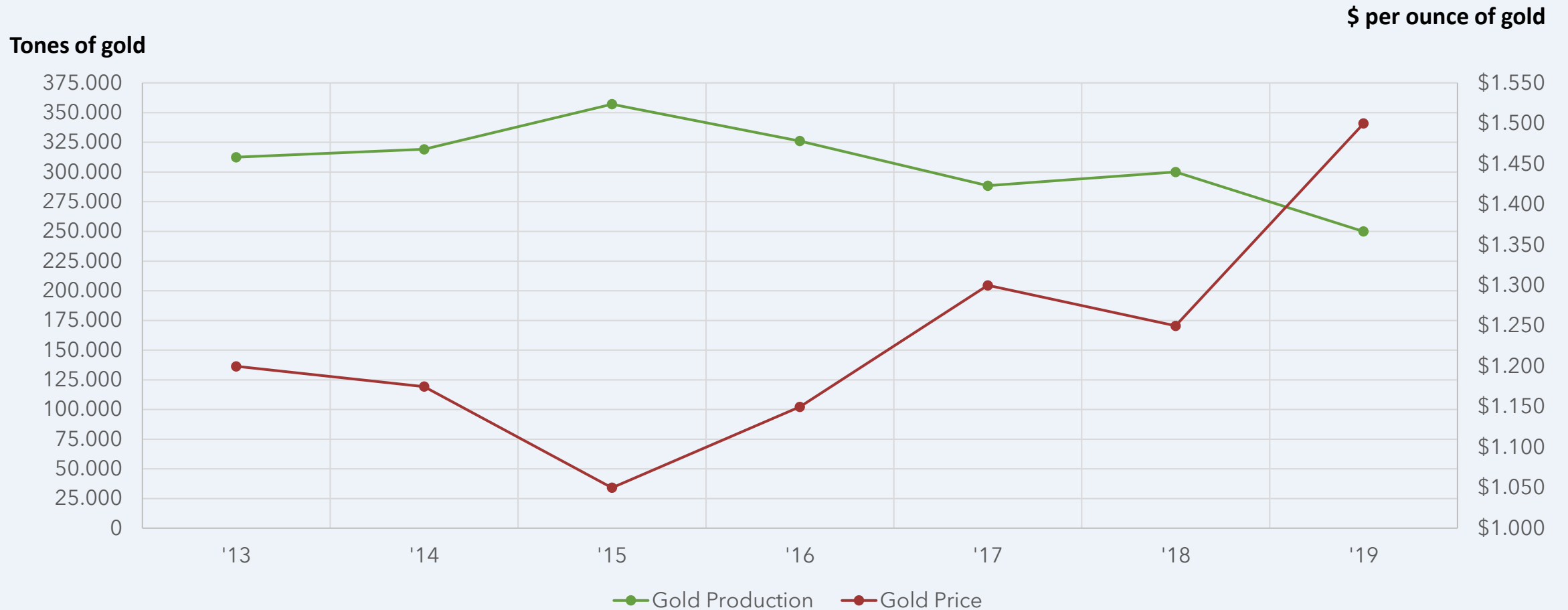
In this section, the candidate should brainstorm any type of costs related to mining and prioritize which one should be focused on. Many structures could be appropriate here, variable/fixed, or by cost item (salary, maintenance, overhead, etc.) The interviewer should try to keep the interviewee focused on mining costs. The interviewee should also come up with a priority list and explain why. No prior knowledge of the mining industry is required here. It is the same principle as any big industrial process i.e., Airlines, Manufacturing, etc.

**Once this is finished, the interviewee should be given Exhibit 2**

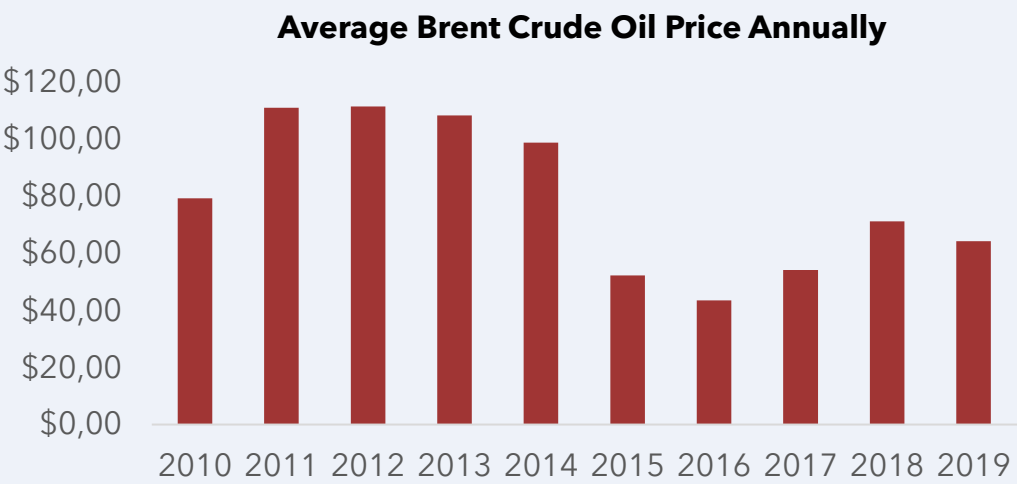
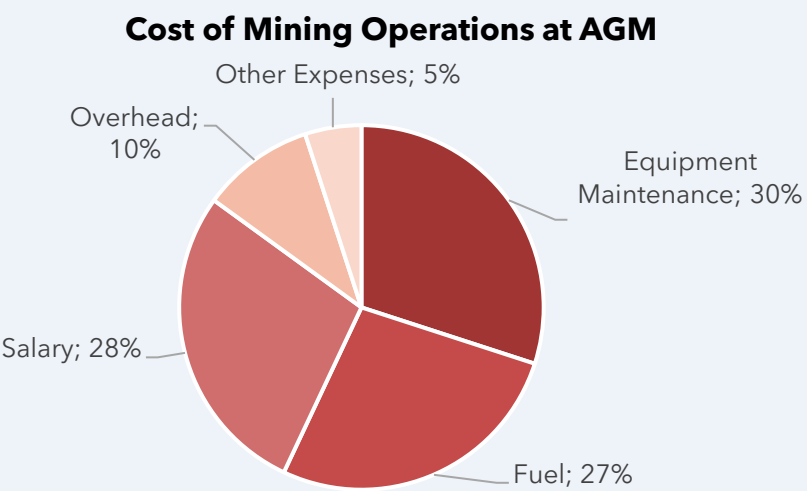
# African Gold Mines Co.

Mining ..... Easy  
Operations ..... Medium  
Profitability ..... Hard

## EXHIBIT 1 - REVENUE STRUCTURE OF AGM



## EXHIBIT 2 - COST STRUCTURE OF AGM MINING DEPARTMENT



| COSTS VARIATION YoY AT AGM MINING OPERATIONS |     |     |     |     |     |
|----------------------------------------------|-----|-----|-----|-----|-----|
| Costs                                        | '15 | '16 | '17 | '18 | '19 |
| Salary                                       | 1%  | 0%  | -2% | -1% | 1%  |
| Maintenance                                  | 3%  | 9%  | 9%  | 12% | 19% |
| Fuel                                         | 3%  | 3%  | 5%  | 9%  | 13% |
| Overhead                                     | 1%  | 1%  | 2%  | 0%  | 1%  |
| Other                                        | 1%  | 5%  | 5%  | 0%  | 0%  |

# African Gold Mines Co.

---

Mining ..... Easy  
Operations ..... Medium  
Profitability ..... **Hard**

## TAKEAWAYS – EXHIBIT 2

---

Exhibit 2 is purposely very busy to evaluate how the candidate can associate information from different sources together.

In the end, the candidate should quickly identify maintenance costs and fuel costs as the two main drivers of the drop in profitability of AGM, but also acknowledge that salary is a big cost.

**Once the candidate has indicated s/he would like to investigate maintenance/fuel, hand over Exhibit 3.**

## TRUCK SELECTION ALTERNATIVES ANALYSIS

---

**Our client has been looking at replacing its fleet of mining trucks as the fleet is getting very close to the end of its useful life. They have done some analysis and found four possible alternatives. Show Exhibit 3.**

## TAKEAWAYS – EXHIBIT 3

---

The interviewee should compare the financials of the four alternatives. The exhibit also gives the state of the current situation for comparison purposes. The four alternatives are:

1. Overhaul of the current fleet by keeping the same trucks but changing major components.
2. Buying a new fleet of the exact same truck.
3. Buying a fleet of a new model of diesel trucks from a different supplier.
4. Buying a fleet of new semi-electric autonomous trucks from the same supplier that require less fuel and no driver to operate.

## CALCULATION – EXHIBIT 3

| ALTERNATIVES                                 | FUEL COST PER YEAR                 | MAINTENANCE PER YEAR | CAPEX PER YEAR     | TOTAL COST PER YEAR         |
|----------------------------------------------|------------------------------------|----------------------|--------------------|-----------------------------|
| Current Situation<br>Caterpillar 777E        | $20 \times 300 \times 95 = \$570k$ | \$540k               | \$100k             | $570 + 540 + 100 = \$1210k$ |
| Overhaul of Current<br>Caterpillar 777E      | $20 \times 300 \times 80 = \$480k$ | \$500k               | $600/5 = \$120k$   | $480 + 500 + 120 = \$1100k$ |
| New Caterpillar 777E                         | $20 \times 300 \times 60 = \$360k$ | \$340k               | $1000/10 = \$100k$ | $360 + 340 + 100 = \$800k$  |
| Komatsu HD785-7                              | $20 \times 300 \times 70 = \$420k$ | \$450k               | $1200/12 = \$100k$ | $420 + 450 + 100 = \$970k$  |
| Caterpillar 780D<br>Autonomous Semi-Electric | $20 \times 300 \times 40 = \$240k$ | \$360k               | $1600/8 = \$200k$  | $240 + 360 + 200 = \$800k$  |

## FURTHER INFORMATION (if requested)

- Trucks are working 20 hours per day
- Trucks work 300 days per year
- Price of Fuel is 1\$/L
- The mine uses 20 trucks
- Specification and capacity of all trucks are similar. They all can do the job
- Ignore the time value of money
- Assume straight-line depreciation
- Candidate should also compare the new alternative to the current to see if it fulfills our client requirement of reducing cost by 6m.  
Old situation – New =  $\$1210k - \$800k = \$410k$  per truck x 20 trucks = Savings of 8.2m per year. Regardless of which option the candidate chooses, it meets our client requirement

## EXHIBIT 3 - TRUCK SELECTION ALTERNATIVES FOR AGM

| ALTERNATIVES                                     | CAPEX  | MAINTENANCE COSTS<br>PER YEAR | FUEL CONSUMPTION | EXPECTED LIFE |
|--------------------------------------------------|--------|-------------------------------|------------------|---------------|
| Current Situation:<br>Caterpillar 777E           | \$0.1m | \$540k                        | 95 L/hr          | 1 yr          |
| Overhaul of Current<br>Caterpillar 777E          | \$0.6m | \$500k                        | 80 L/hr          | 5 yrs         |
| New Caterpillar 777E                             | \$1.0m | \$340k                        | 60 L/hr          | 10 yrs        |
| Komatsu HD785-7                                  | \$1.2m | \$450k                        | 70 L/hr          | 12 yrs        |
| Caterpillar 780D<br>Autonomous Semi-<br>Electric | \$1.6m | \$360k                        | 40 L/hr          | 8 yrs         |

## TRUCK SELECTION CRITERIA/RISKS - BRAINSTORMING

From the calculation of the four alternatives, the interviewee should understand that both the new 777E trucks or 780D autonomous trucks are financially the same. **The interviewer should prompt the interviewee to brainstorm on criteria/risks that our client should consider when making the decision.** This brainstorming could be structured in many forms, but Pros & Cons would be appropriate.

### SAMPLE BRAINSTORMING

|                 | PROS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 777E            | <ul style="list-style-type: none"> <li>• Same model the client has been using</li> <li>• Same spare parts / supply chain</li> <li>• Same maintenance processes</li> <li>• Good training of employees</li> <li>• Longer life 10 years</li> </ul>                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Possible higher wage costs</li> <li>• Possible higher operation costs if fuel prices increase</li> <li>• Potentially more dangerous options</li> <li>• Operational efficiency might be less</li> <li>• Require continuous training of operators</li> </ul>                                                                                                                                                                                                                                                               |
| 780D Autonomous | <ul style="list-style-type: none"> <li>• Lower fuel consumption, costs saving could be more important if fuel prices go up</li> <li>• Further cost saving possible if employees can be laid off</li> <li>• Increased safety benefits</li> <li>• Potential increased operational benefits since these trucks could be more efficient</li> <li>• No training required for operators</li> <li>• Lower fuel consumption = Lower GHG emissions, a more environmentally friendly option</li> </ul> | <ul style="list-style-type: none"> <li>• New technology which might be harder to implement in a very remote site in Africa.</li> <li>• Technical support might be harder to obtain</li> <li>• Maintenance and operation personnel will require training, and this might be hard to obtain in a remote setting</li> <li>• New spare parts / supply chain required</li> <li>• Old inventory might not be usable</li> <li>• Laying-off employees in a small village in Africa might not be appreciated by the communities</li> <li>• Shorter life 8 years</li> </ul> |

# African Gold Mines Co.

---

Mining ..... Easy  
Operations ..... Medium  
Profitability ..... **Hard**

## RECOMMENDATION

---

**Our client is coming to our office in a couple of minutes and we need you to make a recommendation.**

## SAMPLE RECOMMENDATION

---

In this case, there is no one good answer. Both the 777E and the 780D options are financially equal and both could be defended. The interviewee should use the result of the financial analysis and the brainstorming to make a sound and logical argument for which option is chosen.

**The recommendation should include:**

- The interviewee should structure the recommendation by stating that the AGM has a cost problem mainly based on the high maintenance cost and fuel consumption of the old trucks.
- To reduce their costs to their previous level, the company should change its fleet of mining trucks.
- Indicate which option s/he chooses and back that up with logical arguments from the brainstorming.
- Acknowledge some of the risks related to that option.

# Nica Productions

---

*By Alfonso Tomás Durandeu (IESE MBA 2021)*



|                       |       |               |
|-----------------------|-------|---------------|
| Media & Entertainment | ..... | Easy          |
| Profitability         | ..... | <b>Medium</b> |
| Operations Cost       | ..... | Hard          |

# Nica Productions

---

Media & Entertainment ..... Easy  
Profitability ..... **Medium**  
Operations Cost ..... Hard

## PROMPT

---

Nica Productions is an American Media company that is trying to figure out its next project. This company has an extensive experience producing series and movies for all types of audiences and has got many awards doing so. This company has two alternatives: to produce a series for a streaming company or a movie to be projected in cinemas worldwide.

Producing media content implies big investments and low certainty about potential incomes, which depends on many factors; for that reason, our client has hired us to help her decide which is the best alternative for her.

## CLARIFYING POINTS (if asked)

---

- There is no specific profitability goal
- The company is known worldwide, with access to top star directors, actors and technical staff
- It has not budget limitation
- Both alternatives look for a worldwide reach but target different type of customers
- Production of any alternative will last one year
- There is no alternative project
- The main source of revenue of both projects depends on audience

## CASE GUIDANCE

---

This is a quantitative case that requires the candidate to estimate the potential cashflow of different alternatives in order to get the NPV and decide the best option for the client.

The candidate will need to ask for additional information that is necessary to solve the problem, rather than relying on the interviewer to dispense it.

Especially for less finance-read candidates, you may have to help nudge through the math and formulae.

## INTERVIEWER GUIDANCE – STRUCTURE

---

The candidate should express that the company will **pursue the project that generates the higher positive profits** and show that in his/her Framework.

- A good candidate will take into consideration uncertainties related to production and revenues streams and express the intention to estimate NPV of each project.
- Other aspects to bear in mind are competition, market trends, company's strategy, etc.

If asked for information about **REVENUES and COST**, make him/her **BRAINSTORM about it**.

- A good candidate would understand that revenues come not only from tickets or broadcasting royalties but also from merchandising, games, DVD/Blu-rays, etc. In terms of cost, the candidate should mention the basics: director, actors, production, marketing, etc.

## PROJECT ALTERNATIVES ANALYSIS – MOVIE

---

The company is not sure about **which type of movie they want to produce and if it will be able to hire the director and actors for the desired alternative**. However it was able to assign probabilities for each scenario. Income will depend on the size of audience movie attracts, which further depends on the critics received. **Which project (movie or series) should our client choose? (Show Exhibit 1).**

## ANALYSIS TAKEWAYS - MOVIE

---

In order to estimate the NPV of this alternative, the candidate should **calculate the corresponding Cash Flows and request for a Discount Rate (20%)**. The investment are made at the beginning of the project and incomes are received at the end of year 1 (Solution in the following slide).

### A good candidate will:

- read the note to get the information about how much money the company receives from Cinema chains
- be structured and present the calculation with clarity (ideally, in the form of a decision tree)

## ANALYSIS CALCULATION - MOVIE

### TOTAL AUDIENCE (TA):

$$TA = [ (AudienceGood * \%Good) + (AudienceBad * \%Bad) ]$$

$$TA \text{ NOLAN} = [ (600M * 70\%) + (400M * 30\%) ] = [ 420M + 120M ] = 540M$$

$$TA \text{ BAY} = [ (400M * 80\%) + (200M * 20\%) ] = [ 320M + 40M ] = 360M$$

### INCOME:

#### 1ST YEAR (Year 0)

There is no income

#### 2ND YEAR (Year 1)

$$\text{Income} = \text{Total Audience} * \text{Ticket Price} * \% \text{Commission}$$

$$\text{Income NOLAN} = 540M * 10 \text{ USD/t} * 10\% = 540M \text{ USD}$$

$$\text{Income BAY} = 360M * 10 \text{ USD/t} * 10\% = 360M \text{ USD}$$

### INVESTMENT:

Each alternative has its own cost structure (See exhibit 2)

|                  | ALTERNATIVE 1 |     | ALTERNATIVE 2 |     |
|------------------|---------------|-----|---------------|-----|
|                  | GOOD          | BAD | GOOD          | BAD |
| Probability (%)  | 70%           | 30% | 80%           | 20% |
| Audience (M)     | 600           | 400 | 400           | 200 |
| Contribution (M) | 420           | 120 | 320           | 40  |
| Average audience | 540           |     | 360           |     |

| (M USD)                | ALTERNATIVE 1 |     | ALTERNATIVE 2 |     |
|------------------------|---------------|-----|---------------|-----|
| Year                   | 0             | 1   | 0             | 1   |
| Investment             | -150          | 0   | -100          | 0   |
| Income                 | 0             | 540 | 0             | 360 |
| Cash Flow              | -150          | 540 | -100          | 360 |
| NPV                    | -150          | 450 | -100          | 300 |
| <b>Accumulated NPV</b> | <b>300</b>    |     | <b>200</b>    |     |

|                            | ALTERNATIVE 1 | ALTERNATIVE 2 |
|----------------------------|---------------|---------------|
| Probability (%)            | 60%           | 40%           |
| NPV (M USD)                | 300           | 200           |
| Contribution (M)           | 180           | 80            |
| <b>Average NPV (M USD)</b> | <b>260</b>    | -             |

## EXHIBIT 1: MOVIES ALTERNATIVES

|                 | Alternative 1                       |         | Alternative 2                       |         |
|-----------------|-------------------------------------|---------|-------------------------------------|---------|
| Target Audience | Teenagers and Young Adults          |         | Kids and Teenagers                  |         |
| Probability (%) | 60%                                 |         | 40%                                 |         |
| Director        | Christopher Nolan                   | 25M USD | Michael Bay                         | 10M USD |
| Main Actor      | Tom Hardy                           | 50M USD | Mark Wahlberg                       | 25M USD |
| Support Actor   | Michael Cain                        | 25M USD | Tyrese Gibson                       | 5M USD  |
| Others          | Production & Marketing              | 50M USD | Production & Marketing              | 60M USD |
|                 | Critics                             |         | Critics                             |         |
| Good            | 70% chance of getting 600M audience |         | 80% chance of getting 400M audience |         |
| Bad             | 30% chance of getting 400M audience |         | 20% chance of getting 200M audience |         |

*Note: Filmmakers receive as income 10% of the ticket sales (10 USD/ticket);*

## PROJECT ALTERNATIVES ANALYSIS - SERIES

The company already has the **script for a potential Series**, and in case of choosing this alternative, it will receive an upfront payment from the Streaming company. Income will be based on the Audience per episode in the first year (**Show Exhibit 2**).

## ANALYSIS TAKEWAYS - SERIES

In order to estimate the NPV of this alternative, the candidate should **calculate the corresponding Cash Flows and request for a Discount Rate (20%)**. The investment is made at the beginning of the project and incomes are received at the end of year 1 (Solution in the following slide)

### A good candidate will:

- understand that the Script expense (10M USD) is a sunk cost and it should not be taken into consideration to make the decision
- be structured and present the calculation with clarity (ideally, in the form of a decision tree)

**If the candidate realizes about the sunk cost, she/he will choose to produce a Series. If not, make him/her identify the mistake**

|        | NPV | NPV (with sunk cost) |
|--------|-----|----------------------|
| Movies | 260 | 260                  |
| Series | 270 | 260                  |

## ANALYSIS CALCULATION - SERIES

**TOTAL AUDIENCE (TA):**

$$TA = [ (AudienceGood * \%Good ) + (AudienceMedium * \%Medium) + (AudienceBad * \%Bad) ] * \#Episodes$$

$$TA = [ (100M * 30\%) + (70M * 40\%) + (40M * 30\%) ] * 10 = [ 30M + 28M + 12M ] * 10 = 700 M$$

**INCOME:**

**1ST YEAR (Year 0)**

The income in the first year is the Upfront = 20M USD

**2ND YEAR (Year 1)**

$$Income = Total\ Audience * Income\ per\ episode\ (PMV) = 700M * 600k\ USD = 420M\ USD$$

**INVESTMENT:**

If the company chooses this alternative, it should make an additional Investment of 100M USD  
(The Script is a sunk cost)

| ADDITIONAL DATA                |         |
|--------------------------------|---------|
| Upfront (M USD)                | 20      |
| # episodes                     | 10      |
| Income per episode (PMV) (USD) | 600,000 |

| INVESTMENT (USD)       |     |
|------------------------|-----|
| Cost (M USD)           | 100 |
| Upfront income (M USD) | 20  |
| Investment (M USD)     | 80  |

|                  | GOOD | MEDIUM | BAD |
|------------------|------|--------|-----|
| Probability (%)  | 30%  | 40%    | 30% |
| Audience (M)     | 100  | 70     | 40  |
| Contribution (M) | 30   | 28     | 12  |

| (M USD)         | SERIES |     |
|-----------------|--------|-----|
| Year            | 0      | 1   |
| Investment      | --100  | 0   |
| Income          | 20     | 420 |
| Cash Flow       | --80   | 420 |
| NPV             | --80   | 350 |
| Accumulated NPV | 270    | -   |

## EXHIBIT 2: SERIES

### EXPENSES

|                         |    |
|-------------------------|----|
| Script (already bought) | 10 |
| Actors                  | 60 |
| Production              | 40 |

### ADDITIONAL DATA

|                          |         |
|--------------------------|---------|
| Upfront income (M USD)   | 20      |
| # episodes               | 10      |
| Income per episode (PMV) | 600,000 |

| AUDIENCE LEVEL | AUDIENCE PER EPISODE (M USD) | PROBABILITY |
|----------------|------------------------------|-------------|
| High           | 100                          | 30%         |
| Medium         | 70                           | 40%         |
| Bad            | 40                           | 30%         |

*PMV: per million views*

## PROJECT ALTERNATIVES ANALYSIS – ADDITIONAL INFO

Would your decision change with this new information? (Show Exhibit 3)

### TAKEWAYS – EXHIBIT 3

Show the candidate Exhibit 3, which contains an additional cash flow for Michael Bay movie from merchandising and other incomes.

The candidate should **identify the tendency in the cashflow (CAGR -10%) and calculate a perpetuity.**

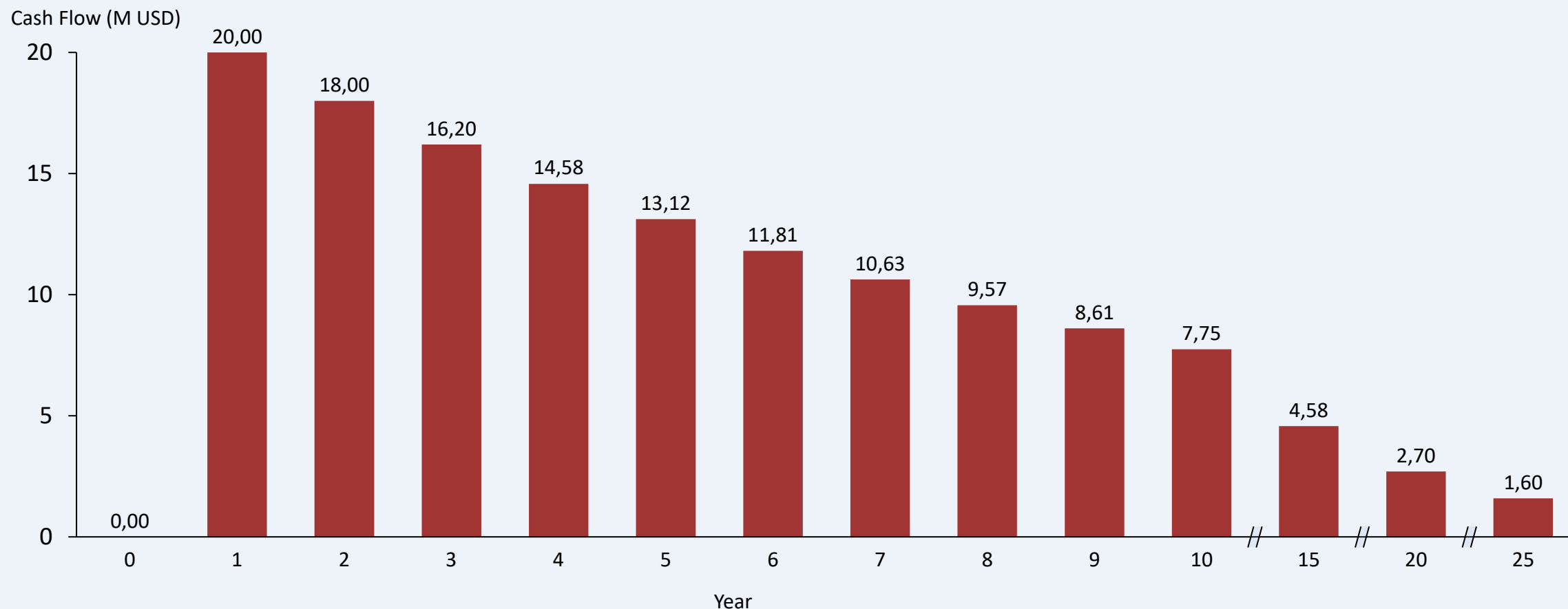
Adding those Cash Flows to Alternative 2 will affect the NPV.

**As a result of this information, the final decision will change.**

|                           | PERPETUITY  |
|---------------------------|-------------|
| Initial Cash Flow (M USD) | 20          |
| Growth (%)                | -10%        |
| Discount Rate (%)         | 20%         |
| <b>NPV (M US)</b>         | <b>66.7</b> |

|                           | ALTERNATIVE 1 | ALTERNATIVE 2 |
|---------------------------|---------------|---------------|
| Probability (%)           | 60%           | 40%           |
| NPV (M USD)               | 300           | 266.6         |
| Contribution (M)          | 180           | 107           |
| <b>Average NPV (M US)</b> | <b>287</b>    | <b>-</b>      |

## EXHIBIT 3 – MOVIES ALTERNATIVE 2



## PROJECT ALTERNATIVES ANALYSIS – OTHER FACTORS

---

**Which other factor would you take into consideration?**

### EXPECTED CONSIDERATION

---

**The candidate should quickly identify the uncertainty and our capability to assess it as the main risk.**

**Factors that can influence:**

- Production is not finished on time and target launch is missed
- The launch of a good movie made by a competitor
- An economic crisis that affects the consumption
- Others

This question is to test the candidate's business sense and creativity. There is room for the candidate to discuss other factors, including other revenue streams and intangible factors, always justifying his/her answer.

# Nica Productions

---

Media & Entertainment ..... Easy  
Profitability ..... **Medium**  
Operations Cost ..... Hard

## RECOMMENDATION

---

**Great, our client is coming and will request a recommendation.**

## RECOMMENDATION – SUGGESTION

---

The candidate should be concise and structured, without mentioning topics that were not discussed.

It is important to highlight the uncertainty of the decision process and he/she should suggest potential ways to reduce it.

# Pipeline Oil Technology

---

*By Roberto Carlos De Araujo (IESE MBA 2021)*



|            |       |               |
|------------|-------|---------------|
| Oil & Gas  | ..... | Easy          |
| Operations | ..... | <b>Medium</b> |
| Pricing    | ..... | Hard          |

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... Medium  
Pricing ..... Hard

## PROMPT

Minerva's University Fluids Research Lab has discovered a more efficient way to transport crude petroleum oil inside pipelines. This new technology can be used in midstream applications where the oil is acquired from the extraction plant and delivered to the refinery plant. The university invested \$ 1,200M in this project during the last 12 years.

The new technology mixes water and oil under certain conditions to reduce the loss of energy, caused by the friction between the oil and the pipeline surface while being transported. As a result, the transport between two given points gets 15% faster and the useful lifetime of the pipelines increases by 20%.

**Minerva's University asked our help to determine the value at which they should sell the technology.**

## CLARIFYING POINTS (if asked)

**What is the market?**

- Mexico.

**How big is the market?**

- 4,800 km of pipeline.

**Who are the competitors and market share?**

- National Oil Company (NOC) is the only player. However, the market is open for the last two years.

**Which are the potential buyers?**

- Primarily, NOC. However, the other two prospects are interested in entering the market.

**Does the University have a patent? How long does it last?**

- The University has already filed for a patent, which lasts for 20 years.

**What is NOC pipelines current capacity?**

- Full capacity. Surplus is transported by more expensive means

such as rail car, barge and truck.

**How is the demand for crude oil?**

- NOC sells all the crude oil it buys. See Exhibit 2 for the next years' forecast.

**Can NOC build more pipelines to substitute other means of transportation?**

- Yes, it is an alternative. However, there are costs involved. See "4. Given Data - Alternative: Expand Pipeline Network"

**How long does it take to implement this technology?**

- Minerva's University estimates that the technology would be running in 100% of the pipelines in one year at \$2,000 M installation cost.

## CASE GUIDANCE

**Crude oil Value chain**

**Downstream:**

- Exploration - Extraction

**Midstream:**

- Transportation

**Upstream:**

- Refinery - Marketing/Sales

**Midstream Business Model**

1) Buy from the extractors (\*)

2) Transport crude oil from the extractor to the refinery

3) Sell to the refineries at \$10 per barrel

(\*) Assume that Cost of transportation already incorporates the buying price from the extractors.

## INTERVIEWER GUIDANCE – STRUCTURE

Identify the main drivers of the pricing and evaluate the results by comparing to other investments.

### A) PRICING:

Value = (1) Savings Costs of transportation + (2) Savings Cost of replacing pipelines – (3) Cost to implement

#### (1) Savings in Cost of transportation

- Increase in flow speed leads to increase in the pipeline's delivery capacity
- Save costs of sales by switching from other means of transportation to pipelines

#### (2) Savings in Cost of replacing pipelines

- Save Costs by using the pipelines for longer, reducing the replacement

#### (3) Cost of installation

- \$2,000 M

### B) COMPARISON: Compare the value of the technology to other investments to validate the final pricing

#### (4) Research Investment (by Minerva's University):

- \$1,200 M (spent in the last 12 years by Minerva's University)

#### (5) Alternative investment (for National Oil Company):

- Investment to expand the pipeline network from 2 M to 2.5 M barrels per day

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## REVENUE ANALYSIS

**Hand over Exhibit 1** if the candidate asks for revenues and/or costs breakdown

## DEMAND ANALYSIS

**Hand over Exhibit 2** if the candidate asks about the demand for the next years

## PRICING CALCULATION

**(1) Savings in cost of transportations (SCOT):**

SCOT = (I) Sales for 20 years \* (II) Savings in switching transportation = \$162,000 M \* 0.035 = \$5,670 M ~ \$6,000 M in 20 years (or \$300 M / year)

(I) Sales for 20 years = Chart area (trapezoid) \* 360 days \* price per barrel: =  $[(2+2.5) * (20) \times (1/2)] * 360 * 10 = \$162,000 \text{ M}$

(II) Savings in switching transportation = % increase pipeline use \* proportional savings = 10% \* 0.35 = 0.035

Switching from truck (5%) and rail (5%) to pipeline: % increase pipeline use = 15% \* 70% = 10.5% ~ 10%

Proportional Savings:= [% truck \* (cost truck – cost pipeline) + % rail \* (cost rail – cost pipeline)] = 5% \* (\$6 - \$2) + 5% \* (\$5 - \$2) = 0.35

## TIP: ROUND OFF NUMBERS

Suggest the candidate to round off numbers to facilitate easier calculations. Some suggestions are underlined in the case. Nonetheless, strong candidates figure out these opportunities by themselves.

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## EXHIBIT 1

**Income Statement**

|                               |           |
|-------------------------------|-----------|
| + Sales                       | \$7,200 M |
| - Cost of transportation (*)  | \$1,980 M |
| - Cost of replacing pipelines | \$1,000 M |
| - Other operating costs (**)  | \$3,500 M |
| Operating profit              | \$720 M   |

**Table 1:** Income statement for Year 0, assuming sales of 2M barrels per day.

| Means    | Cost           | Barrels |
|----------|----------------|---------|
| Pipeline | \$2 per barrel | 70%     |
| Truck    | \$6 per barrel | 5%      |
| Barge    | \$3 per barrel | 10%     |
| Rail     | \$5 per barrel | 15%     |

**Table 2:** Cost of Sales breakdown per means of transportation

(\*) Cost of transportation already includes the buying price from the extractors.

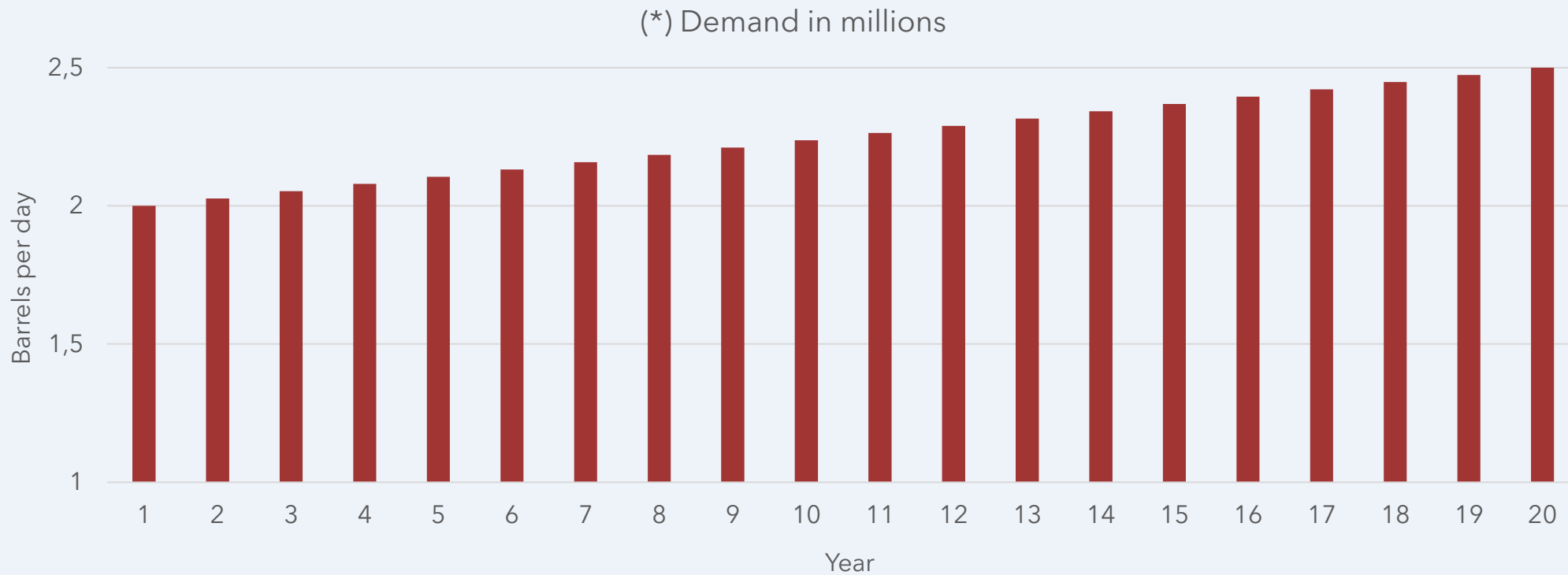
(\*\*) Assume that Other Operating Costs do not change over the years.

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## EXHIBIT 2

### Crude oil demand forecast



(\*) In a given year, assume that the demand per day is the same for all the 360 days.

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## PRICING CALCULATION

### (2) Savings in Cost of replacing pipelines (SCRIP):

= Cost of replacing pipelines per year \* 20 years \* (1 - new lifetime/current lifetime)  
= \$1,000M \* 20 \* [1 - 4/5 years] = \$4,000M in 20 years (or \$200M per year)

## ALTERNATIVE ANALYSIS

### Expand network pipeline

#### Investment to expand the pipeline network:

= (Additional capacity in 20 years / Additional Capacity) \* cost to implement  
= (2.5 M – 2M) / 0.05 M \* \$700 M = \$7,000 M

## GIVEN DATA:

- Cost of replacing pipelines: \$1,000M per year
- Lifetime of regular pipeline: 5 years

## GIVEN DATA:

- Additional capacity: 50,000 barrels per day
- Time to implement: 2 years
- Cost per additional capacity = \$ 700 M

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## COMPARISON ANALYSIS

| PRICING                                    | \$8,000 M | INVESTMENTS                                       |           |
|--------------------------------------------|-----------|---------------------------------------------------|-----------|
| (+) Savings in Cost of transportation      | \$6,000 M | (-) Research Investment (University)              | \$1,200 M |
| (+) Savings in Cost of replacing pipelines | \$4,000 M | (-) Alternative investment (National Oil Company) | \$7,000 M |
| (-) Cost of installation                   | \$2,000 M |                                                   |           |

## NEGOTIATION

The University can recover its investment by selling the technology for any price over \$1,200 M. Given the estimated savings and cost of installation, NOC will pay less than \$ 8,000 M to guarantee profits/savings.

Alternatively, NOC can construct its own pipeline network for a total investment of \$7,000 M. Then, it is better to buy the technology for \$ 7,000 or less than to go forth with this alternative investment. Note that the alternative investment is limited by additional 50,000 barrels/day capacity in two years, while the new technology can be put in operation in just one year.

**Therefore, a reasonable price for selling this technology would be between \$1,200 M and \$7,000 M.**

# Pipeline Oil Technology

Oil & Gas ..... Easy  
Operations ..... **Medium**  
Pricing ..... Hard

## RECOMMENDATION

**What would be your final recommendation to Minerva's University?**

## SAMPLE RECOMMENDATION

The general recommendation is open. One of the possibilities is to sell the new technology for a price of \$1,500 M plus a 30% participation in the additional revenue while the University holds the patent (40% of \$10,000M = \$3,000M in 20 years, disregarding cost of installation). There are three reasons that support this proposal:

- 1) Cost savings by increasing the volume of crude oil transported in pipelines. According to the calculations, \$300 M per year (around 40% of the current Operating Profits)
- 2) The improvement in the pipeline lifetime is also relevant accounting for \$200 M per year (around 25% to 30% of the current Operating Profits)
- 3) The alternative of expanding the pipeline network is a higher investment than the cost savings generated by the new technology. Besides that, expanding the pipelines is limited by additional 50,000 barrels/day in two years, while the new technology can be put in operation in just one year.

**A great candidate would also briefly discuss any risks or next steps:**

### **Main risks / sensitive assumptions:**

- Delay in installation of the new technology
- Limitation in reaching some regions, since it can be done only by a specific means of transportation
- High investment (\$2,000M). Options: cash surplus, bank loan, increase in equity
- Crude oil price fluctuation -> Use future contracts to guarantee buying and selling prices
- Demand fluctuations because of crisis or other external factor

### **Next steps:**

- Verify calculations with NOC's calculations/data to validate the assumptions
- Define a negotiation strategy based on the calculations/assumptions
- If the negotiation fails, look for other prospective buyers

# The Bookstore

---

*By Víctor Manzanares Bonilla (IESE MBA 2021)*



|              |       |               |
|--------------|-------|---------------|
| Retail       | ..... | Easy          |
| E-commerce   | ..... | <b>Medium</b> |
| Market Entry | ..... | Hard          |

# The Bookstore

---

Retail ..... Easy  
E-commerce ..... **Medium**  
Market Entry ..... Hard

## PROMPT

---

Our client, Classic Bookstore (CB), is a traditional bookstore chain in Spain, specialized in non-technical books. CB's revenue has stagnated for the past 3 years, with a stable and loyal customer base. Now, as part of its new growth strategy, CB is considering whether to enter the electronic books market.

The client is considering to sell a CB branded reading device and develop a website to sell e-books for it.

**Our client asked us to analyze this opportunity and provide a recommendation.**

## CLARIFYING POINTS (if asked)

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- CB only sells to customers through physical stores, no online business is available.
- CB has stores in the largest cities in Spain.
- CB has 3 types of customers: AVID READERS (2 books/month), OCCASIONAL READERS (1 book/2 months) and RARE READERS (1 book/6 months).
- Non-technical physical book market in Spain has been stagnated for the past 3 years.
- CB has reached an agreement with an e-Reader manufacturer in China. Total cost per device would be 60€. These devices can only support the e-books sold on CB's new website.
- CB has no specific growth rate in mind and are open to suggestions from us.

## CASE GUIDANCE

---

This is a case designed to be led by the candidate. Start by reading the case question and let the candidate drive the analysis. Do not provide any information until it is asked.

This case primarily tests the understanding of market entry and its implications on the current business model.

For simplicity, taxes and value of money over time have been ignored in this case, although excellent candidates should mention them during the case.

## INTERVIEWER GUIDANCE – STRUCTURE

---

Candidate's structure should cover the following key aspects of the problem.

**MARKET OPPORTUNITY**: What is the **market size** of generic e-books in Spain? What is this market's **growth**?

**POTENTIAL SHARE**: What would be our **market share**? How many competitors are we facing in this market?

**POTENTIAL PROFIT**: What is the potential **profit** of this new market? Expected revenues vs expected costs? What **investment** is required to enter in this new market? What is the expected **return on investment** of our client? Payback period?

**CAPABILITIES & RISKS**: Does this new market align with our client's **strategy** and **capabilities**? Do they have the know-how required? Have they got the financial capabilities to undertake this investment? What is the potential **cannibalization** of this new business model with the current one?

The analysis should be led by the candidate, starting for the market size. When the candidate requests information about the market and size, ask him to estimate the size of the non-technical books in Spain, both in paper and e-books.

## INTERVIEWER GUIDANCE – MARKET SIZING

(Suggested approach)

1. Population of Spain: 45M

2. Target population that reads: We assume people from 15 to 80 years old.

Population 0 – 20 (25%): 11.25M (Population 15-20:  $11.25/4=2.8$ M)

Population 21 - 40 (25%): 11.25M

Population 41 – 60 (25%): 11.25M

Population 61 – 80 (25%): 11.25M

Total target population =  $2.8 + 11.25 + 11.25 + 11.25 = 36.5$  M

3. Percentage of population that buys books: We assume 60% of people between 15 and 80.

Total target population =  $36.5\text{M} \times 0.60 = 22\text{M}$  people

4. Type of readers: We assume the following categories and quantities

Avid Readers: 12 books/year  $\times$  2M readers = 24M books/year

Occasional Readers: 6 books/year  $\times$  5M readers = 30M books/year

Rare Readers: 2 books/year  $\times$  15M readers = 30M books/year

Total books:  $24 + 30 + 30 = 84\text{M}$  books

5. Percentage of paper books and e-books and average prices:

PAPER BOOKS (93% of books):  $78\text{M books} \times 15\text{€}/\text{book} = 1,170\text{M€}$

E-BOOKS (7% of books):  $6\text{M books} \times 8\text{€}/\text{book} = 48\text{M€}$

**TOTAL MARKET:  $1,170 + 48 = 1,220\text{M €}$**

# The Bookstore

---

Retail ..... Easy  
E-commerce ..... **Medium**  
Market Entry ..... Hard

## PROFITABILITY ANALYSIS – GIVEN DATA

---

In the next step, the candidate should focus on profitability. **Hand Exhibit 1** for this part and **provide the following information if requested:**

### REVENUE

- Market Annual Growth: 5%
- Expected Market Share: 1%
- Paper book Gross Margin: 33%
- E-book Gross Margin: 40%
- E-reader price: To be determined by the candidate with **Exhibit 2**. Price range should be between 60 and 100€ to compete against Kindle.
- No customers change from paper to e-book after year 1. New readers are coming from new customers.

### COSTS

- Webpage Investment: 150,000€
- General Expenses website: 50,000€
- Cannibalization: Candidate needs to calculate the number of users that will switch from paper to e-reader with **Exhibits 1 & 3**

# The Bookstore

Retail ..... Easy  
E-commerce ..... Medium  
Market Entry ..... Hard

## PROFITABILITY CALCULATION

| YEAR                  | 1                                | 2               | 3               | 4               | 5               |
|-----------------------|----------------------------------|-----------------|-----------------|-----------------|-----------------|
| Market Size           | 48M €                            | 50.4M €         | 52.92M €        | 55.6M €         | 58.38M €        |
| Revenue e-books       | 480,000 €                        | 504,000 €       | 530,000 €       | 560,000 €       | 584,000 €       |
| Gross Margin          | 190,000 €                        | 201,600 €       | 212,000 €       | 224,000 €       | 232,000 €       |
| Profit from e-readers | 120,000 € *                      | 3,000 € **      | 3,250 €         | 3,750 €         | 3,000 €         |
| Webpage investment    | - 150,000 €                      | -               | -               | -               | -               |
| General Expenses      | -50,000 €                        | -50,000 €       | -50,000 €       | -50,000 €       | -50,000 €       |
| Cannibalization       | -90,000 € (50,000 books x -1.8€) | -90,000 €       | -90,000 €       | -90,000 €       | -90,000 €       |
| <b>TOTAL PROFIT</b>   | <b>20,000 €</b>                  | <b>64,600 €</b> | <b>75,250 €</b> | <b>87,750 €</b> | <b>95,000 €</b> |

(\*) This calculation has taken into account a gross margin of 10€ per e-reader. Candidate will have to pick the price in each case. 12,000 readers are sold the first year, 11,000 to customers switching from paper to e-book and 1,000 to new customers.

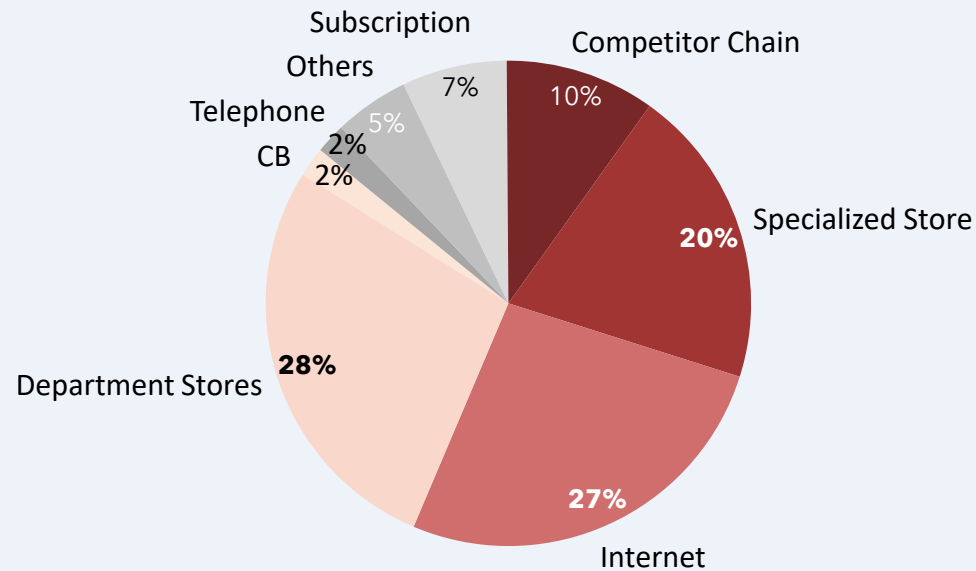
(\*\*) No customers switch from paper to e-book after 1<sup>st</sup> year. New readers are sold to new customers acquired by market growth.

# The Bookstore

Retail ..... Easy  
E-commerce ..... Medium  
Market Entry ..... Hard

## EXHIBIT 1: NON-TECHNICAL BOOKS MARKET IN SPAIN

PAPER BOOKS MARKET IN SPAIN

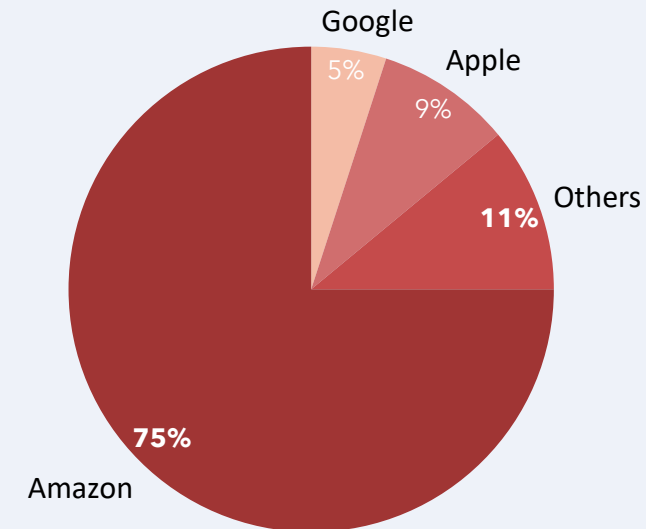


Market Size: 1,170M€

Annual Growth = 0%

Average price per book: 15€

E-BOOKS MARKET IN SPAIN



Market Size: 48M€

Annual Growth = 5%

Average price per e-book: 8€

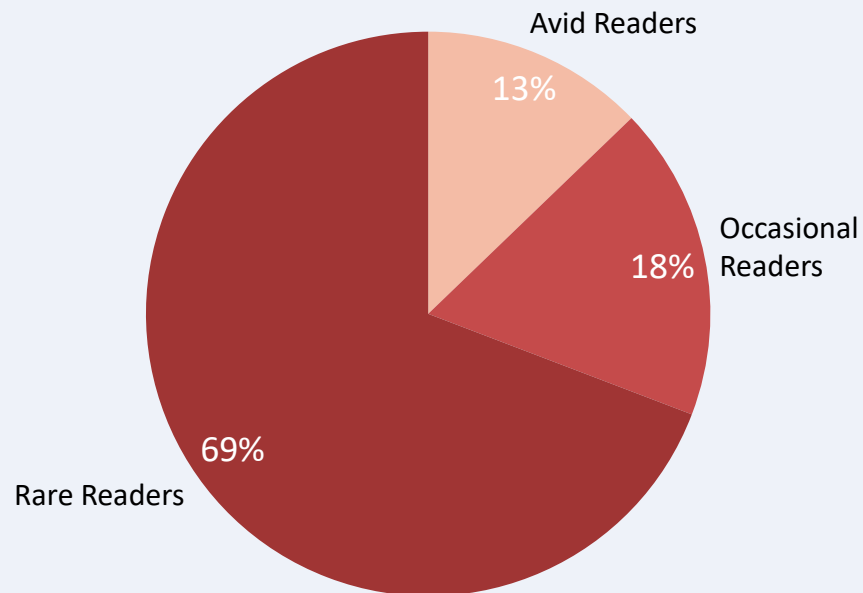
Average consumption per reader: 10 e-books/year

## EXHIBIT 2: ELECTRONIC READERS

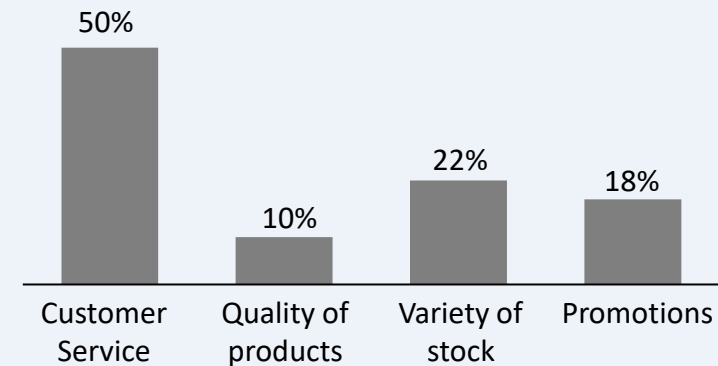
|                  | <b>CB e-Reader</b> | <b>Sony e-Reader</b> | <b>Amazon Kindle</b> | <b>Amazon Kindle PRO</b> |
|------------------|--------------------|----------------------|----------------------|--------------------------|
| Price            | TBD                | 120€                 | 100€                 | 150€                     |
| Cost             | 60€                | -                    | -                    | -                        |
| Formats accepted | CB                 | All formats          | All formats          | All formats              |
| Storage          | Up to 1,000 books  | Up to 1,500 books    | Up to 750 books      | Up to 2,000 books        |
| Extra Features   | Medium             | Medium               | Low                  | High                     |

## EXHIBIT 3: CB'S CUSTOMERS INFORMATION

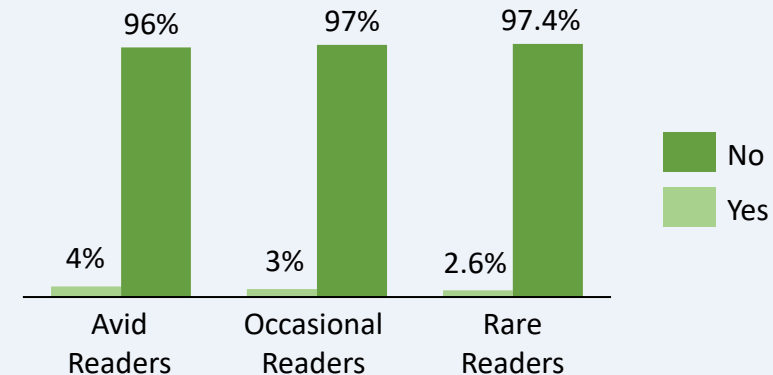
TOTAL CUSTOMERS: 390,000



What do you value the most about CB?



Would you consider switching to an electronic reader?



# The Bookstore

---

Retail ..... Easy  
E-commerce ..... Medium  
Market Entry ..... Hard

## BRAINSTORMING 1

---

How would you launch this product?

### EXPECTED BRAINSTORMING 1

---

In this question, the analysis should be carried out by focusing on the following main aspects:

- **Segmentation:** What users are we targeting? What could be our main target considering the company's strategy and client base?
- **Product:** How can we highlight the strong aspects of our product? What do the customers want and how can we meet their demands?
- **Price:** Although the price has been set before, additional measures can be explored, such as promotions, free gifts to great customers, etc.
- **Promotion:** How should this product be marketed? What promotions should be used?
- **Place:** Through which channels should this reader be sold?

## BRAINSTORMING 2

---

What other measure could our client implement in order to increase revenues?

### EXPECTED BRAINSTORMING 2

---

In this question, the candidate should come up with additional measures to increase the current revenue streams. This measures should include:

- Increase number of products. Maybe including technical books in our offer could increase the number of customers.
- Creation of a loyalty scheme to try to increase average spending per customer.
- Creation a referral program to increase our customer base.
- Negotiate with e-reader supplier to include additional features.

# The Bookstore

---

Retail ..... Easy  
E-commerce ..... **Medium**  
Market Entry ..... Hard

## RECOMMENDATION

---

**What is your final recommendation for Classic Bookstore?**

## SAMPLE RECOMMENDATION

---

- The candidate should recommend to enter in this new market of electronic readers and e-books.
- Based on our calculations and projections, our client should expect a return of 342,600€ over 5 years from an initial investment of 150,000€.
- RISKS: 1) Calculations have been based on projections of market share and a survey given by the client. Any deviation from this data could affect the profitability of the investment. 2) Similar book chains to CB could enter this market and reduce our client's potential market share. 3) Cannibalization with our client's current business model could damage the company's results and image.
- POTENTIAL NEXT STEPS: These risks could be mitigated by producing a deeper market analysis and carrying out further surveys among customers to have a more accurate prediction of the market behavior. Different programs could be explored to increase customer loyalty and new customers acquisition.

**Excellent candidate:**

- A candidate who points out that taxes and time value of money have not been considered during the profitability analysis and this would reduce the profitability of the investment.
- A candidate who mentions the lack of experience and knowledge of our client in this new business as part of the risks. This could lead to a reduction in customer service, which is very valued by our CB's clients.

# Gymco

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*By Pieter Swart (IESE MBA 2021)*



Sports ..... Easy  
Wellbeing ..... **Medium**  
Growth Strategy ..... Hard

## PROMPT

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Your client is an international chain of fitness centers, operating in Sub-Saharan Africa, Europe and Southeast Asia

GymCo missed its 2013 growth target of ZAR600M

**The CEO would like you to investigate what is going on**

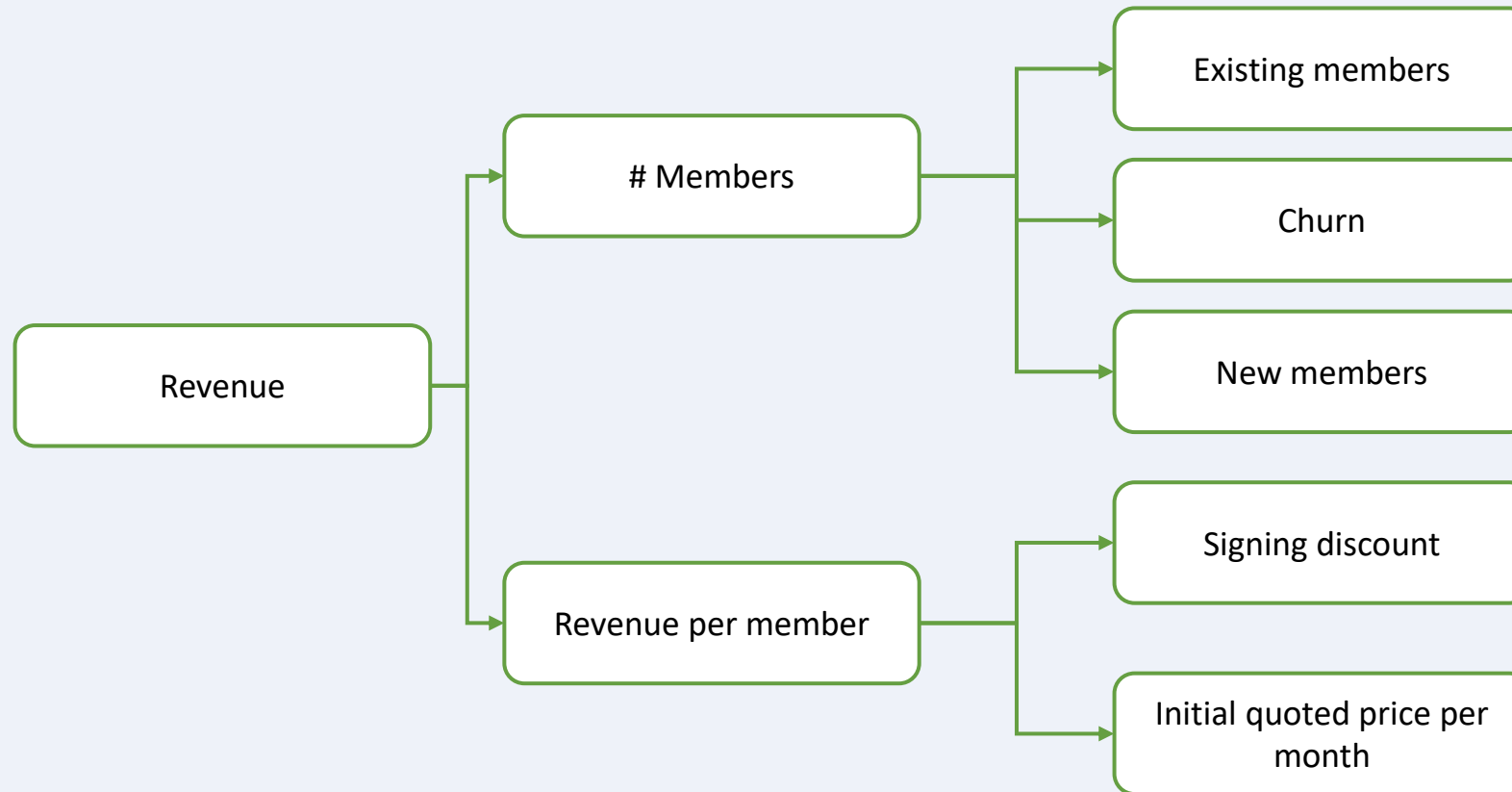
## CLARIFYING POINTS (if asked)

---

- There are 2 major gym chains, GymCo has 60% market share, FitnessCo has 30%, and a few small chains the remaining 10%
- GymCo members pay a monthly membership fee of ~ZAR700 pm
- Market trends are in favour of gyms – consumers are switching to have more healthy habits
- There are a few smaller competitors that have recently entered the market – these are smaller gyms offering more classes, with less focus on free weights and cardio sections
- No other competitors have noticed any decline in revenues; in fact, they have had strong increases over the past 12 months

# INTERVIEWER GUIDANCE – STRUCTURE

## Structure example



## REVENUE AND/OR VOLUME ANALYSIS

---

**Hand over Exhibit 1** if the interviewee asks for revenues and/or members volume

## EXPECTED TAKEWAYS FROM EXHIBIT

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- # members is increasing, but revenue is not increasing by the same rate – therefore revenue per member must be decreasing
- I see that churn and # of new members jump up as revenue starts declining (early 2013), did a lot of people leave and rejoin for some reason? I'd like to explore that more
- The overall takeaway is that it seems that we offered some sort of discount package that caused more people to join / rejoin

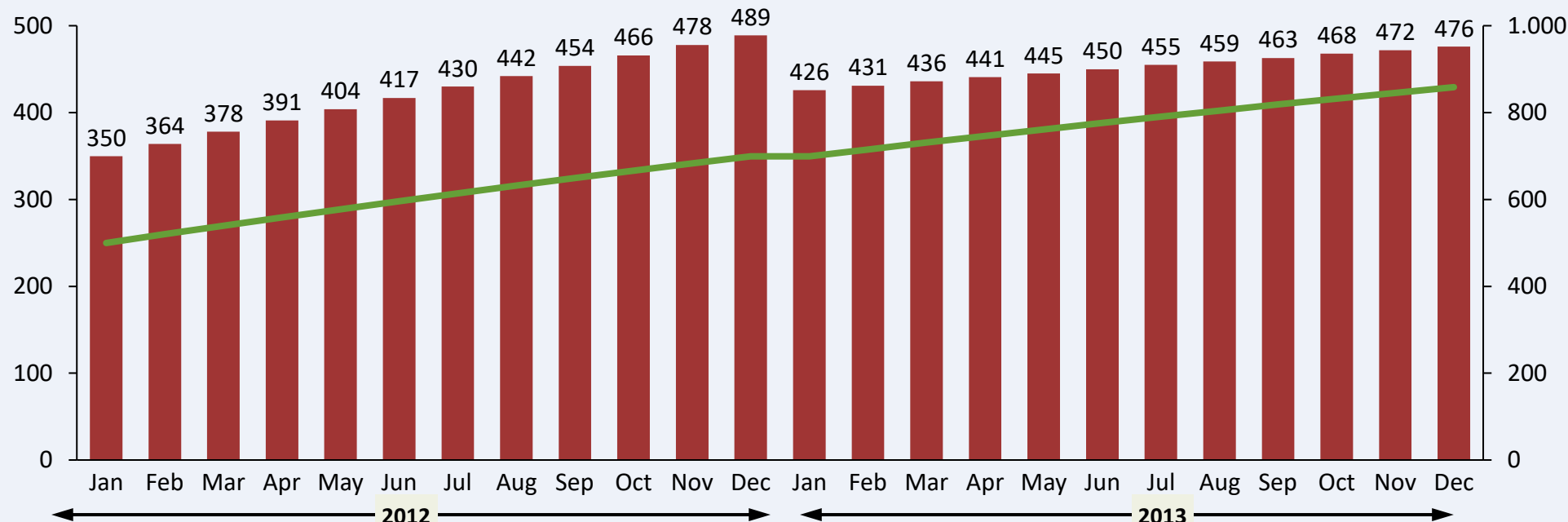
## FURTHER INFO FOR CANDIDATE (if asked)

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- Yes, indeed – GymCo partnered with a big health insurer (HealthCo) in 2012, where they would offer 40% off to HealthCo members – the intention was to increase sign-ups (Normal price ZAR700 pm, HealthCo price ZAR400 pm)
- However, during 2013, GymCo management started noticing that some members were leaving the gym, switching healthcare providers to HealthCo, and rejoining at the new discounted rate; the estimation is that 50% of people who left the gym during 2013, rejoined at the lower rate
- Effectively, GymCo was now making less revenue from its existing members, and only some new revenue from new sign-ups

## EXHIBIT 1: REVENUE AND #MEMBERS OVER TIME

— #Members ('000) ■ Revenue (ZAR M)



# new members ('000) – end of month

30 30 30 30 30 30 30 30 30 30 30 205 30 30 30 30 30 30 30 30 30 30 30

Churn rate (at end of month)

2% 2% 2% 2% 2% 2% 2% 2% 2% 2% 2% 30% 2% 2% 2% 2% 2% 2% 2% 2% 2% 2% 2%

## ASSESSMENT OF CONTRACT ANALYSIS

---

The GymCo CEO wants to know the real financial impact of the contract – has it been beneficial? If not, how much has GymCo lost as a result?

## EXPECTED CONSIDERATION

---

**Use this to prompt candidate and frame calculation**

In order to assess the decision, we need to consider the incremental revenues from the contract

**Additional revenues:**

- # Brand new members who joined because of the discount, and would not have joined without the discount

**Foregone revenues:**

- # of existing members who switched to the HealthCo discount (i.e. who would have paid R700, but now only pay R400)
- # of new members who joined on HealthCo discount who would have joined at full price irrespective of the discount

**Provide Exhibit 2 when the candidate points out the above considerations. If they do not explain the above, explain it to them and then provide them with Exhibit 2**

## EXHIBIT 2

| 2013 figures - All figures in '000              | 2013 |
|-------------------------------------------------|------|
| # Normal members at start of year               | 699  |
| # Normal members joining during year            | 60   |
| # Normal members leaving during year            | 313  |
| # Normal members at end of year                 | 446  |
| # HealthCo Discount members at start of year    | 0    |
| # New HealthCo discount members during year     | 475  |
| # HealthCo discount members leaving during year | 66   |
| # HealthCo Discount members at end of year      | 409  |

## CALCULATION INFO

- Normal members pay ZAR700 pm, HealthCo Discount members pay ZAR400 pm
- At the start of 2013, 200K normal members left and rejoined with HealthCo discount (i.e. they would be willing to pay the full price, but rejoined to get the R400 discount)
- Of the brand new sign-ups, 30% would have joined GymCo at full price anyway (irrespective of the discount, i.e. they would have paid ZAR700pm instead of the discounted ZAR400pm)

## CALCULATION

### How much revenue is GymCo foregoing from existing members that are switching providers?

We need to segment the # of new HealthCo members into three groups –

- **Group 1:** those that were existing members who rejoined,
- **Group 2:** Those who would have joined anyway, regardless of discount
- **Group 3:** Those that are completely new members, joining as a result of discount

**Group 1:** # Existing members who rejoined = 200K;

- lost revenue =  $200K (700-400) \text{ pm} * 12 \text{ months (since they all rejoined at the start of the year)} = \text{ZAR}720\text{M}$  revenue foregone

**Group 2:** # of HealthCo members who would have joined GymCo irrespective of the HealthCo deal =  $475K - 200K \text{ (existing GymCo members)} = 275K * 30\% = 82.5K$

- Revenue foregone on these members who would have joined anyway =  $82.5K * (700-400) * 6 \text{ months (average duration, assuming joining throughout the year)} = \text{ZAR}297\text{M}$  revenue foregone

**Group 3:** # of members acquired as a result of HealthCo deal =  $475K - 200K \text{ (existing GymCo members)} - 82.5K = 192.5K$  new members

- Additional revenue =  $192.5 * \text{ZAR}400 \text{ per month (HealthCo rate)} * 6 \text{ months (average duration, assuming joining throughout the year)} = \text{ZAR}462\text{M}$

Therefore additional revenue of ZAR 462M minus foregone revenue of ZAR1017M (297M + 720M) = **net negative effect of 555M**

**The contract is very detrimental to GymCo**

### Expected Insight:

Furthermore, the impact in 2013 seems to have been even greater, and this new contract with HealthCo is putting a strain on GymCo's business model – too much reliance on HealthCo for new members, unable to get new members organically

Therefore, GymCo should try and renegotiate or cancel the contract

# WHAT SHOULD GYMCO DO?

What are the options available to GymCo with regards to the contract?

## EXAMPLE RESPONSE

GymCo has a few options available:

- Get out of the deal with HealthCo
- Reduce the discount that HealthCo is giving members
- Change policies to prevent members from rejoining for a certain time period (e.g. 12 months) if they have left the gym

| Option                                                                    | Benefits                                                                                                                   | Drawbacks                                                                                                                                                                                 |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Get out of the deal                                                       | No loss in revenue due to no more switching                                                                                | <ul style="list-style-type: none"><li>• Potential loss in future # of clients</li><li>• What to do about existing discounted members? If we charge full price, they might leave</li></ul> |
| Reduce discount                                                           | <ul style="list-style-type: none"><li>Discourage switching</li><li>Still incentive for brand new members to join</li></ul> | <ul style="list-style-type: none"><li>• Likely reduced # of new members</li><li>• What to do about existing discounted members? If we change their price, they might leave</li></ul>      |
| Change policy                                                             | No impact on # of new sign-ups                                                                                             | <ul style="list-style-type: none"><li>• Long-term, perverse incentive still exists</li></ul>                                                                                              |
| Change to limited offer (i.e. discount only lasts for the first 6 months) | Likely to get the same number of new sign-ups                                                                              | <ul style="list-style-type: none"><li>• Discounted clients might leave at the end of the discounted period</li></ul>                                                                      |

## RECOMMENDATION

---

The CEO wants to meet with us in a few minutes to discuss our findings as well as the way forward – what will you tell him?

## SAMPLE RECOMMENDATION

---

- GymCo is currently in a very onerous contract with HealthCo, and it should cancel it. The contract was having a negative impact of ZAR555M per annum
- Furthermore, this contract places too much reliance on and gives too much power to HealthCo
- GymCo should find a way to get new members organically, while retaining existing HealthCo members
  - Offering discounts for yearly subscriptions
  - Marketing ideas
  - Offering competing classes (to compete with the smaller gyms)
  - Promotions on new members (bags / towels etc.)
- In order to decide on the best strategy, I would like to quantify the above options (cost vs. benefit)

# Green Airlines

---

*By Antonio Niemeyer (IESE MBA 2021)*



|                     |             |
|---------------------|-------------|
| Airlines            | Easy        |
| Growth Strategy     | Medium      |
| Investment Decision | <b>Hard</b> |

# Green Airlines

Airlines ..... Easy  
Growth Strategy ..... Medium  
Investment Decision ..... **Hard**

## PROMPT

Due to the recent bankruptcy of a major airline, the aviation authority of Brazil recently opened an auction for landing and takeoff slots in one of the country's biggest airports. A slot is the right to land and depart from an airport during a given time period.

Green Airlines, a small, regional airline operating in the North part of the country, was offered 10 slots, for the total price of \$100M. If Green accepts to buy the slots, it will have to operate them for at least 5 years.

**The owner and CEO of Green Airlines approached your firm looking for an advice on whether they should buy the slots or not.**

## CLARIFYING POINTS (if asked)

- An airport slot is a permission granted by the owner of an airport, which allows the grantee to schedule a landing or departure at that airport during a specific time period
- Green airlines currently does not operate in the airport that is being discussed
- Green airlines currently only flies regional flights and has no plans to include international flights in its offerings
- Green airlines currently does not have the necessary planes to operate the slot. Management will need to lease 5 airplanes to operate the 10 slots
- The \$100M that Green Airlines would have to pay is a one-off payment, due before operations start
- The main objective of the owner/CEO is financial gain
- Green Airlines can sell the slots, but only after five years of operation
- If Green Airlines does not buy the slots, they will be sold to another airline

## CASE GUIDANCE

In this case, the candidate will need to lead through directive questioning. Although the answer may seem straightforward, the case will force the candidate to analyze the problem from different perspectives; it will demand not only math and problem-solving skills, but also, creativity.

A strong candidate will quickly realize that in order to answer the questions of Green Airlines' CEO, it will be necessary to understand the strategic fit and the financial implications of buying the slots. After realizing that it does not make sense to buy the slots to operate them, the candidate should explore alternative way to explore the opportunity that has emerged.

## INTERVIEWER GUIDANCE – STRUCTURE

---

The candidate should realize that this is an opportunity for Green Airlines to expand its business into one of the country's main airport.

Suggested items-to-consider are:

**Market** - What's the trend for the demand of flights in the airport's region? What's the profile of travelers (business or leisure)? Are the other airlines going through financial difficulties? Is the industry suffering in general or was the bankruptcy a one-off event?

**Competition** - How many companies operate at this airport? Are the slots currently concentrated in the hands of a few companies or are they split among several companies? Do Low Cost Carriers operate in the airport? What's competitor's price?

**Revenues/Costs** - What's the expected number of passengers per day per slot (plane size, load factor, flights/day)? What's the expected price per passenger? What are the fixed costs? What are the variable costs?

**Internal Capabilities** - Does Green Airlines have the operational capabilities necessary to operate the slots (planes, overhead, sales system, suppliers)? Does operating in a big airport demand a different strategy than operating small, regional airports? Does Green Airlines have the financial capabilities necessary to pay for the slots? If not, what are its options?

**Risks/Alternatives** - Cultural issues of setting up operations in a different area. Are there other regions that might be more attractive?

## INTERVIEWER GUIDANCE — PART 1

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If the candidate raises concerns related to **competition, operational challenges or the aviation market in Brazil**, hand **Exhibit 1** (next page) to clarify these points.

## KEY TAKEAWAYS — EXHIBIT 1

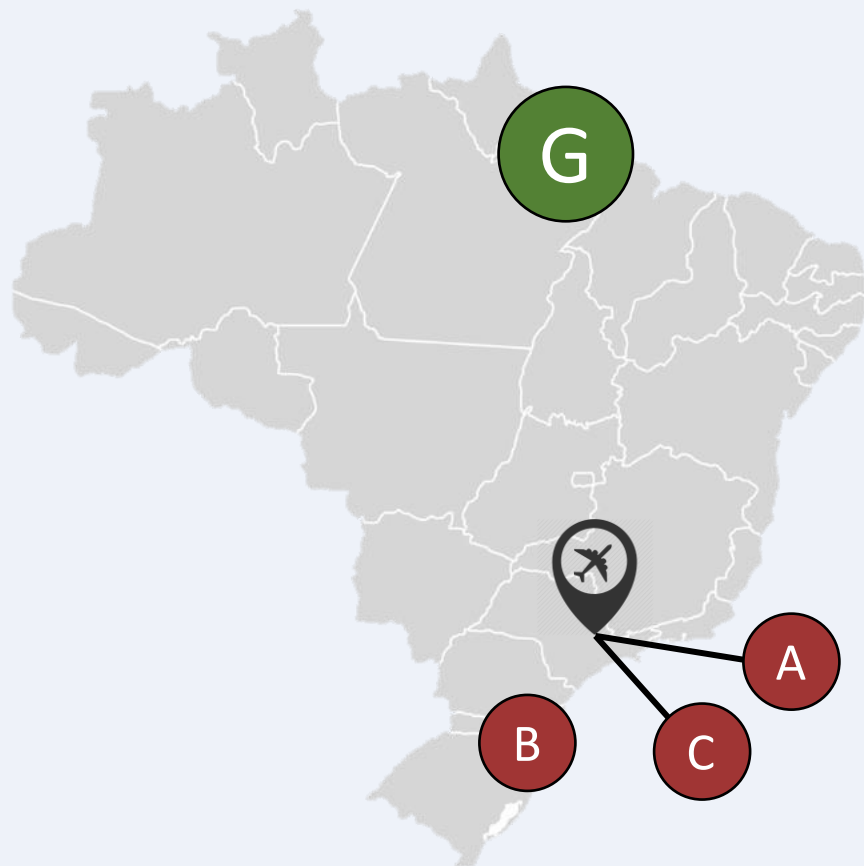
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- The candidate should notice that Green Airlines operates in a very distinct region of Brazil and is much smaller than the main players of the Sao Paulo region
- Buying the 5 planes would mean almost doubling Green Airlines fleet, and represents a significant strategic shift
- Airlines A and C are big players located in Sao Paulo, and they currently operate planes bigger than the other airlines, however, the price per ticket is lower (this may be due to shorter flights and high competition)
- The average load factor in Sao Paulo is significantly higher than the average for Green Airlines
- If Green Airlines were to buy the slots, it would most likely need to have bigger planes and keep prices low

# Green Airlines

Airlines ..... Easy  
Growth Strategy ..... Medium  
Investment Decision ..... **Hard**

## EXHIBIT 1



| Current Operations       | Airlines A | Airlines B     | Airlines C | Green Airlines |
|--------------------------|------------|----------------|------------|----------------|
| City Hub                 | São Paulo  | Santa Catarina | São Paulo  | Pará           |
| Total # of planes        | 100        | 80             | 70         | 6              |
| Avg plane size (# seats) | 300        | 200            | 300        | 100            |
| Avg Load Factor (%)      | 80%        | 70%            | 80%        | 60%            |
| Avg Ticket Price (\$)    | 200        | 250            | 200        | 300            |

-  Airport selling slots
-  Airlines A Main Hub
-  Airlines B Main Hub
-  Airlines C Main Hub
-  Green Airlines Main Hub

## REVENUE ANALYSIS

First step is to estimate the potential revenues of the slot operation. **Ask the candidate what factors s/he would use to estimate the revenues.** When asked, provide the following information in the table:

### Revenues - For operation of 10 slots

|                         |            |
|-------------------------|------------|
| # of planes             | 5          |
| flights / plane / month | 50 flights |
| Seats / plane           | 250 seats  |
| Average Load Factor (%) | 80%        |
| Average Ticket Price    | \$ 200     |

## REVENUE CALCULATION

$$\begin{aligned}\text{Revenue per month} &= \# \text{ of planes} * \# \text{ flights/plane} * \# \text{ seats} * \text{load factor} * \text{ticket price} \\ &= 5 * 50 * 250 * 80\% * 200 = 50,000 \text{ passengers} * \$200/\text{passenger} \\ &= \text{\$10 million/month}\end{aligned}$$

## COST ANALYSIS

Second, the candidate should estimate the costs of operating the slots. **Ask her/him what s/he believes to be the main costs of an airline (fuel, crew, maintenance, insurance, leasing, fees, overhead, etc.).** After discussing the main lines of cost, provide the following information in the table:

### Costs - for operation of 10 slots

|                          |                             |
|--------------------------|-----------------------------|
| Initial Investment       | \$5M (to set up operations) |
| Fuel                     | \$14k per flight            |
| Other Variable Costs     | \$4k per flight             |
| Salaries                 | \$1,5M per month            |
| Maintenance & Leasing    | \$400k per plane/month      |
| Insurance, Fees & Others | \$2M per month              |

## COST CALCULATION

Fuel:  $\$14k * 50 * 5 = \$ 3,5M/month$

Other Variable Costs:  $\$4k * 50 * 5 = \$ 1,0M/month$

Salaries:  $\$ 1,5M/month$

Maintenance & Leasing:  $\$400k * 5 = \$ 2,0M/month$

Insurance, Fees & Others:  $\$ 2,0M/month$

**Total Cost: \$10M/month**

**Expected Profit:**

**Year 1:**  $-\$5M + \$120M - \$120M = -\$5M$

**Following years:**  $\$120M - \$120M = \$0$

**Expected Insight:**

The candidate should realize that the expected **operational result is -\$5M in Year 1, and zero in the following years** for the 10 slots on sale.

## ALTERNATIVE ANALYSIS

---

After concluding that the operating profit would be zero for the slots, **ask the candidate what additional analyses s/he would make in order to decide whether to buy the slot or not.**

## POTENTIAL ALTERNATIVES

---

The candidate should come up with potential alternatives:

### **Improve operational metrics**

- Possibility to increase revenues (increase ticket price, include non-ticket revenues, offer packages, shuttle services, etc.)
- Possibility to reduce costs (use bigger planes to reduce fixed costs, negotiate lease terms, exclude food inflight, automatization of processes, change fuel supplier, etc.)

### **Buy slots and sell to other company**

- Airlines A and Airlines C have their Hubs in Sao Paulo. The slots are probably worth a lot for them.
- How much is the market value of a Slot?

## INTERVIEWER GUIDANCE – PART 2

After discussing the potential alternatives, **state that management has already explored all alternative ways to improve the operational result, and the numbers presented are already considering all operational improvements possible.**

## EXPECTED TAKEWAYS

If the candidate does not reach this solution by herself/himself, **say that the slots are very valuable for the big airlines operating in the region.**

The big airlines have operational advantages related to **scale**. They operate bigger planes (300 seats) than Green Airlines, so their potential revenues are higher (all other assumptions remain the same, including costs).

**Ask the candidate to calculate the value of the 10 slots for the big airlines, considering planes with 300 seats.**

**Expected Calculation:**

**New Revenue per month** =  $5 * 50 * 300 * 80\% * 200$  = **\$12M/month**

**New Monthly profit** = \$2 million/month  
= **\$24 million/year**

## INTERVIEWER GUIDANCE – PART 2

Now that we know the operational results for a big company utilizing these 10 slots, **ask the candidate to estimate the value of the slot for a big company such as Airlines**

**A.** Provide the following information if requested

### Valuation of Slots

|               |              |
|---------------|--------------|
| Discount Rate | 10% per year |
| Right to use  | perpetual    |

## CALCULATION

**Expected calculation (assuming the Net Profit as a perpetuity):**

Value of slots =  $\$24\text{M} / 10\% = \mathbf{\$240\text{M}}$

**Expected Insight:**

The candidate should identify that the 10 slots have a total value of approx. \$ 240M for the big airlines, and that the best choice is to buy the slots, operate them for 5 years at zero profit, and sell them to a big airlines for a value between \$105M and \$240M.

## SAMPLE RECOMMENDATION

The recommendation should be that Green Airlines buy the 10 slots, operate them for the mandatory 5 years and then sell them at a potential profit of approx. \$135M

## RECOMMENDATION

**Ask the candidate her/his final recommendation**

# SPECIAL THANKS

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